Open Development of AI Drug Discovery

A Living Map of Methods, Models, Data, Benchmarks, and Process Knowledge

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Chapter 1

Scope and boundary

1.1 Why this report exists

Artificial intelligence for drug discovery is not one task and it is not one model class. It is a chain of decisions that begins with biological mechanism and ends, within the boundary of this project, at preclinical candidate nomination. Along that chain, researchers select assays, reconcile chemical and biological identities, define prediction targets, choose experimental units, construct splits, estimate uncertainty, and decide which computational result deserves laboratory follow-up. A technically impressive model can be irrelevant if any preceding decision is wrong.

AIDD Commons therefore joins four artifacts. **Learn** supplies an 18-track graduate curriculum. **Atlas** records methods, models, datasets, benchmarks, software, standards, and tutorials with claim-level evidence. **Labs** make selected protocols executable and expose both successful and weak results. This **Report** explains the map, the open-development choices behind it, and the gaps that remain. These artifacts share identifiers and editorial rules so that a lesson can point to a resource record, a lab can implement a lesson's protocol, and a correction can propagate across all three.

The intended readers are graduate students, AI and computational-biology researchers, and engineers. Some arrive with molecular biology but little machine learning, others can build models but have not worked with assays, medicinal chemistry, or biomolecular structures. The curriculum is a graph rather than a single linear course because these readers need different entry routes. Its common spine is scientific. It requires the reader to define the decision, the sample, the endpoint, the baseline, the validation unit, and the evidence before choosing architecture.

Drug research is unusually sensitive to category errors. A molecular target can mean a gene, transcript, protein isoform, protein state, complex, binding site, or pathway node. An "activity" value can refer to different assay formats, concentrations, censoring rules, and physical units. A random compound split can answer an interpolation question while being reported as if it tested a new chemical series. A predicted pose is not evidence of binding, and binding is not evidence of a useful cellular phenotype. These distinctions motivate the report's emphasis on data contracts and stage-specific claims.

Machine learning is already used across discovery and development, but its appropriate role varies with data density, experimental noise, mechanistic knowledge, and the cost of a wrong decision (Vamathevan et al. 2019). Broad expectations should therefore be tempered by the actual decision context and prospective evidence (Bender and Cortés-Ciriano 2021). AIDD Commons teaches where a model can alter a decision, where it merely compresses existing evidence, and what experiment would falsify the resulting claim.

1.2 The four artifacts and how they couple

The four artifacts are separate in authorship and separate in review, and they are coupled by a shared resource identifier. Figure 1 draws the coupling. The lesson tiles of Learn are ordered, the cells of Atlas are individually reviewable, the console of Labs carries a run control, and the ruled page of Report carries a figure. One correction placed on the shared identifier rail reaches every artifact that repeated the claim.

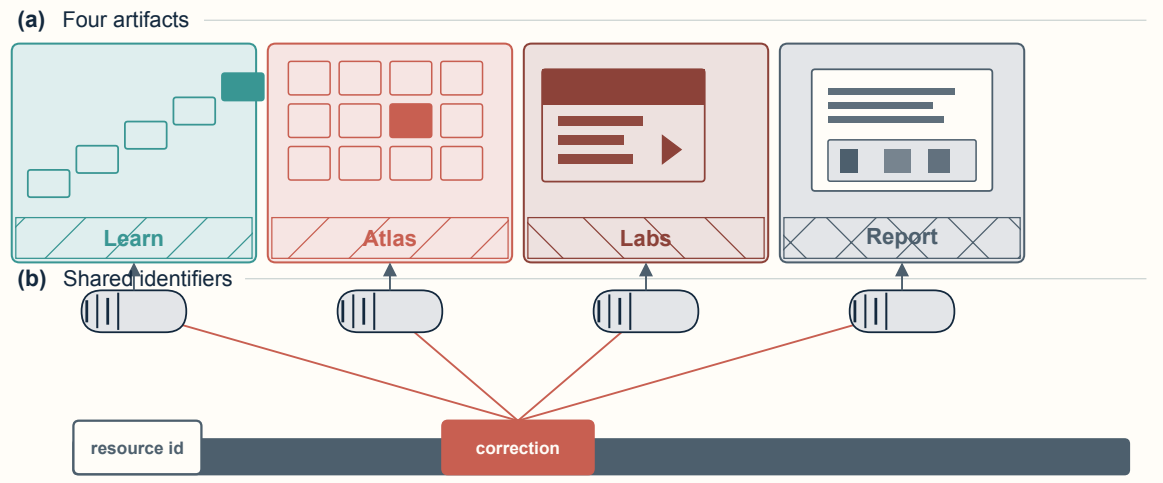


Figure 1: Four artifacts, one system. (a) Each artifact is drawn with its own form. Learn is an ordered staircase of lesson tiles, Atlas a registry of record cells with one cell selected, Labs a console with a run control, and Report a ruled page carrying a figure. The four shapes recur wherever the four artifacts recur, and the four colours are what survive when the colour does not. (b) A rail under the four carries the shared resource identifier. One correction sits on the rail, and the dashed branches carry it to the identifier chip of each artifact, which is the route a correction takes. The figure shows the coupling and nothing more. It carries no counts, no coverage claim and no measured outcome.

The coupling is a property of the record system rather than of the file layout. A resource record holds a stable identifier, a name, a kind, an access level, a lifecycle status, seven openness dimensions with their evidence, a list of licences with their scopes, and a dated list of the pages that were opened. Everything else in the project is generated from that record or refers to it.

Table 1 sets out what each artifact holds and what it cannot establish on its own. The rightmost column matters most, because each artifact is regularly asked a question it was never designed to answer.

Table 1: What each artifact holds and what it does not establish.

Artifact	What it holds	The unit of its content	What it does not establish
Learn	Lessons, exercises, worked cases, and their prerequisites	One learning objective attached to one lesson	That a reader achieved the objective, or that a model performed well
Atlas	A reviewed record of a method, model, dataset, benchmark, standard, or tutorial	One resource with its own evidence	That the resource is open, maintained, or scientifically strong

Artifact	What it holds	The unit of its content	What it does not establish
Labs	A frozen protocol, a command, and a retained result artefact	One protocol with one decision unit and one metric	That the result transfers beyond the frozen split, or that the reading is a result
Report	The map, the argument, the open-development account, and the audit	One claim with its source	That the project is comprehensive, and that a superseding claim is established

The last row is where the report's own limits live, and Chapter 8 states them at length. The third row is where a laboratory result stops. A result opened once under a frozen protocol is evidence about that protocol and about the decision unit it named, and not about a different endpoint, a different chemical series, or a different programme.

1.3 Inclusion boundary

The central scope runs from disease mechanism and target discovery through candidate nomination:

disease definition, mechanism, target identification, and target validation; assay and data strategy, including entity resolution and provenance; hit discovery through experimental screening, virtual screening, repurposing, or design; hit confirmation and series formation; hit-to-lead and lead optimization across potency, selectivity, developability, ADMET, and exposure; modality-specific design for small molecules, proteins, peptides, antibodies, and nucleic acids; synthetic planning, experimental iteration, and design–make–test–analyse loops; and evidence integration for a candidate-nomination decision.

Clinical development, real-world evidence, manufacturing, health economics, and regulatory practice are adjacent but not treated as full professional curricula in v0.1. They appear where they constrain earlier decisions, such as translational endpoints, formulation feasibility, privacy, or the difference between retrospective model performance and clinical utility. Wet-lab protocols are described at the level required to interpret data and design computationally meaningful validation. This report is not a substitute for supervised laboratory, medicinal-chemistry, toxicology, clinical, biosafety, or regulatory training.

The boundary is drawn at the point where a computational claim becomes a human decision that carries cost. Everything upstream of that point can be taught with the machinery in this repository. Everything downstream requires a clinical, manufacturing, or regulatory apparatus that the project does not operate and does not simulate.

1.4 Four kinds of statement

Every important statement belongs to one of four classes, and Table 2 gives the test that keeps each honest. An **observed source fact** is directly established by an opened primary paper, official project page, repository file, licence, standard, dataset record, or measured artifact. A **project decision** is an AIDD Commons choice, such as its 18-track structure, seven-dimensional openness model, or a frozen lab split. An **open question** is material uncertainty the reviewed sources have not settled. **Future work** is an intended addition not yet implemented, measured, or reviewed.

This classification is more than editorial style. It prevents a project's aspiration from being written as a result, a product page from being treated as a scientific validation, and an inaccessible asset from being labelled unavailable. It also makes corrections local, since new evidence can change one openness dimension without rewriting a resource's identity or scientific purpose.

Figure 2 draws the four classes as different in kind rather than in degree. An observed source fact shows the opened document and the link that ties it to a claim. A project decision shows several inputs meeting at one node. An open question shows an object that is not yet defined. Future work shows tiles that are planned and not yet built. The solid border marks the two classes that assert something, and the dashed border the two that do not.

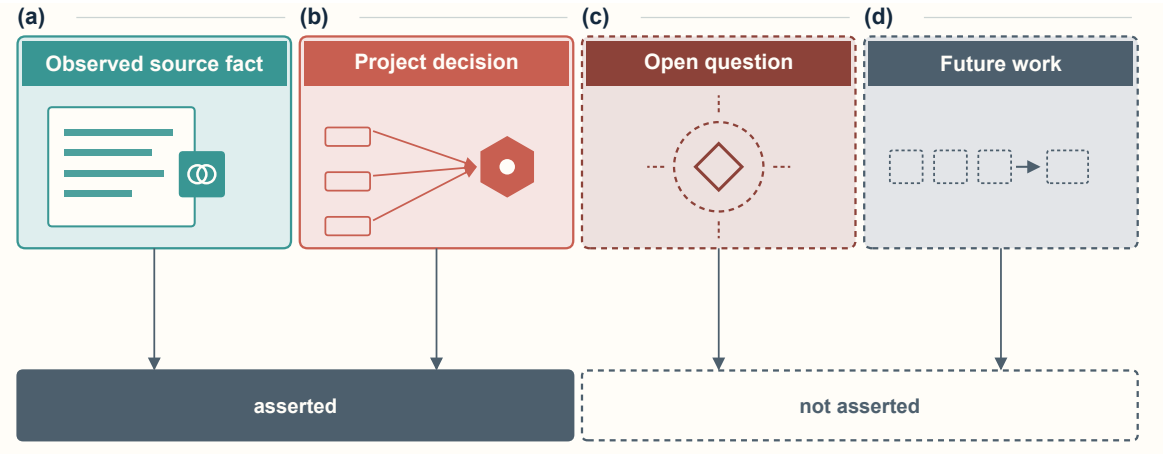


Figure 2: Four classes of statement. The editorial guide makes each statement's authority visible, and the four classes differ in kind rather than in degree, so each is drawn instead of being placed on a scale. (a) An observed source fact shows the opened document and the link that ties it to a claim. (b) A project decision shows several inputs meeting at one node. (c) An open question shows an object that is not yet defined, ringed and spoked. (d) Future work shows tiles that are planned and not yet built. Solid borders mark the two classes that assert something and dashed borders the two that do not, which is the only grouping the guide supports.

Table 2: The four statement classes and the test that keeps each honest.

Class	What it asserts	What may not be inferred from it	Where it is recorded
Observed source fact	A page, file, licence, standard, or measurement says what it says	That the source is correct, complete, or current	The record's evidence list, with a date and a URL
Project decision	The project chose this structure, split, threshold, or rule	That the choice is the only one, or that it is optimal	The protocol, the curriculum registry, or the audit report
Open question	Something material is unsettled by the sources reviewed	That the absence of evidence settles the question	The record's caveats, or the report's limitations
Future work	An addition is intended	That the addition exists, runs, or will work	The roadmap, and only the roadmap

1.5 What v0.1 does and does not claim

This is a living technical report, not a systematic review. The source search is broad and dated, but v0.1 does not claim that every database, paper, course, company, or regional resource has been found. It is not a performance leaderboard. Numerical comparisons appear only when task, data, split, metric, and evaluation protocol are matched closely enough to make the comparison meaningful.

AIDD Commons is **building toward** the most comprehensive open AIDD learning system. It will not promote a superlative coverage claim until a reproducible comparison audit defines the comparison set, units of coverage, evidence threshold, and review date. The present claim is narrower. The project supplies a connected curriculum, an evidence-typed resource registry, a multidimensional openness model, executable honest-baseline labs, and an explicit contribution process in one system.

Chapters 10 and 11 report that audit and the independent reading that checks it, and Chapter 12 reads the frontier families the curriculum now covers. The claim gate that governs the wording of the superlative is described in Chapter 8, where the report states the current status of both candidate claims and the single condition that keeps them from being established.

1.6 Reading the map

Readers should use the artifacts together. Begin with a pathway in Learn, inspect candidate resources in Atlas, and run a lab only after reading its data and evaluation contract. A resource's `last_verified` date describes the evidence review, not the age of the science. `Unverified` is a maintained state, not a negative judgement. Licence fields apply only to their declared asset scope. A benchmark result has no portable meaning outside its protocol.

The rest of this report follows the same logic as a drug-discovery project. Chapter 2 maps decisions from target to candidate. Chapter 3 positions the educational and infrastructure landscape. Chapter 4 asks what open development means when frontier models are only partly inspectable. Chapters 5 and 6 define data and evaluation discipline, and Chapter 6 states the project's own recorded coverage. Chapter 7 describes how a community maintains the map, and Chapter 8 states the safety and licence position, the limits of the present release, and the roadmap. The last four chapters report what the repository holds rather than what it intends. Chapter 9 reports the laboratory summaries the project has committed, Chapter 10 explains how the comparison audit turns reviewed pages into bounds on thirteen audited systems, Chapter 11 reports what an independent reading found, and Chapter 12 reads the eight frontier method families the curriculum now teaches.

Every figure is drawn by script from a committed artefact, so the artwork and the numbers behind it cannot drift apart silently. Each figure names what it shows and what it does not, and the prose beside it says what a reader should look at.

Chapter 2

The target-to-candidate pipeline

2.1 A decision pipeline, not a software pipeline

Drug discovery is often drawn as a funnel. The picture is useful only if each narrowing step states what evidence entered, what was rejected, and what decision followed. Attrition is not automatically failure, because early rejection can be valuable when it prevents an expensive, weakly justified programme. Conversely, a large model that ranks compounds does not create value unless the ranking changes which compounds are made or tested and survives an independent experimental check.

The target landscape itself is heterogeneous. Drug targets span proteins, complexes, nucleic acids, and other biological entities, and the evidence connecting a target to an approved therapeutic is not equivalent to genetic or mechanistic evidence at the start of a new programme (Santos et al. 2017). The pipeline must therefore carry biological identity and causal assumptions forward, rather than reducing the problem to a protein sequence and a potency column.

Figure 3 draws the nine stages as a pipeline with a failure branch at each one, which is what a funnel normally hides. Each cell carries the stage, the decision in question form, the narrowing, and the branch that leaves the narrowing to the failure mode recorded for that stage.

The following map states the primary decision at each stage. "Cheap baseline" means the simplest serious comparator that can expose whether a more complex method has useful headroom.

Stage	Decision	Minimum evidence contract	Strong cheap baseline	Frequent failure
Disease and mechanism	Which process is causally relevant?	Cohort, tissue, perturbation, time, phenotype, confounders	Replicated effect with covariate control	Correlation relabelled as mechanism
Target identification	Which entity is actionable?	Stable IDs, isoform/state, direction, disease context	Curated evidence aggregation	Gene/protein/pathway identities collapsed
Target validation	Does perturbing it change the phenotype?	Intervention, dose, model system, rescue or orthogonal evidence	Held-out perturbation or replicated assay	Same evidence reused for discovery and validation

Stage	Decision	Minimum evidence contract	Strong cheap baseline	Frequent failure
Assay and data strategy	What quantity will guide optimisation?	Protocol, units, censoring, controls, batch, provenance	Transparent assay QC and replicate model	Incompatible endpoints pooled
Hit discovery	Which candidates deserve confirmation?	Search space, target state, selection budget, prospective test	Similarity, docking, or simple supervised ranker	Retrospective enrichment treated as prospective success
Hit confirmation	Is activity real and attributable?	Orthogonal assay, purity, interference checks, concentration response	Rule-based nuisance filters plus retest	Aggregation or assay artefact mistaken for binding
Hit-to-lead	Which series has tractable headroom?	Series identity, potency, selectivity, exposure, synthesis	Matched molecular-pair and medicinal-chemistry review	Optimising one endpoint while losing the programme
Lead optimisation	Which trade-off is worth making?	Multi-parameter objectives, uncertainty, constraints, synthesis cycle	Pareto view plus calibrated per-endpoint models	Scalar score hides unacceptable liabilities
Candidate nomination	Is the package coherent enough to advance?	Integrated efficacy, safety, DMPK, CMC, translational rationale	Predeclared evidence matrix	Model confidence substituted for missing experiments

A target product profile is the formal statement of what a stage is optimising towards, and the systematic review of how it is used in academic research gives the field's own account of the fields that must be filled before optimisation begins (Ibnidris et al. 2024). The independent unit at each stage is the field most often chosen silently, and it most often determines whether a result can be pooled.

Figure 4 draws the same nine stages as forms, one per stage, whose fields are the records that stage cannot be argued without. A filled square at the head of a field marks it as required. The contracts differ from stage to stage, and a stage that borrows another stage's contract has quietly changed what the stage means.

2.2 Disease mechanism and target evidence

Target discovery begins with a disease model, not a model architecture. The computational unit may be a patient, donor, tissue, cell state, perturbation, time point, or spatial niche. These are not interchangeable. Thousands of cells from one donor remain one independent donor for many clinical questions. A perturbation screen with many guide RNAs can support within-screen

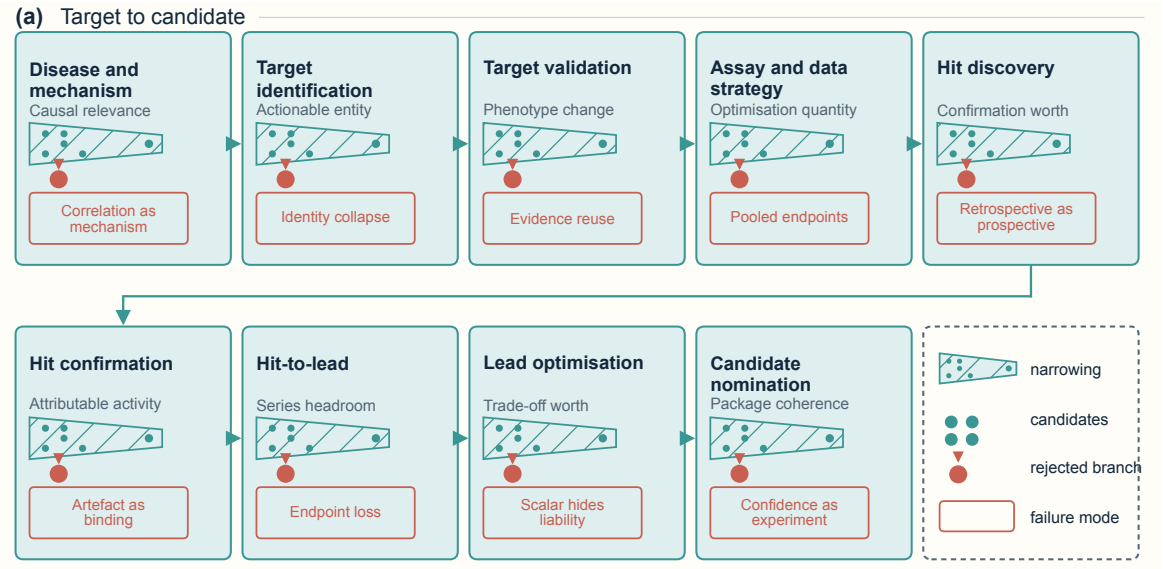


Figure 3: Target to candidate. A stage table of nine stages becomes a pipeline that states what a funnel normally hides. Each cell carries the stage, the decision in question form, a narrowing drawn as a taper with candidates in it, and the branch that leaves the taper to the failure mode the table records for that stage. The route between the two rows gives the order. The dots in a taper are a drawing convention, and the figure reports no count, rate or probability. It also omits the strong cheap baseline that the same table records for every stage.

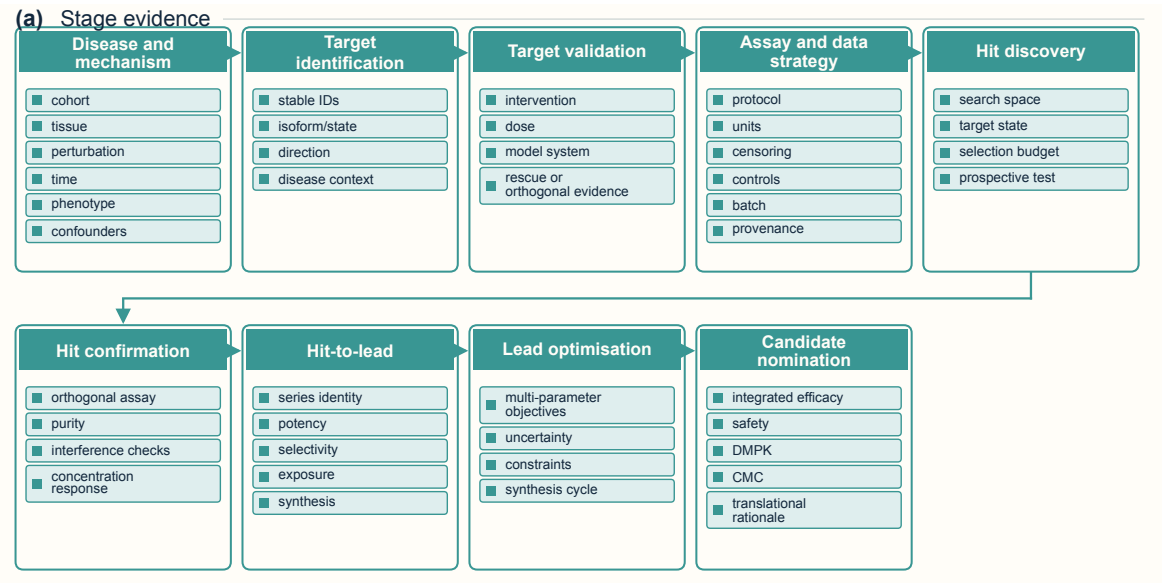


Figure 4: Stage evidence. The same nine stages, each drawn as a form whose fields are the records that stage cannot be argued without, in the wording of the stage table. The filled square at the head of a field marks it as required. The contracts differ from stage to stage, and a stage that borrows another stage's contract has quietly changed what the stage means. The figure shows the evidence contract only. It does not show the strong cheap baseline or the failure modes, which the decision pipeline carries.

learning while offering little evidence for transfer to a different laboratory, background, or disease state.

The first data contract should name the biological entity and its stable identifier, the disease context and the intended direction of modulation, the independent sampling unit, the measurement and its uncertainty, the intervention or natural experiment supporting causality, and the external evidence that could refute the target hypothesis.

Knowledge graphs, language models, omics embeddings, and literature agents can organise evidence, but they inherit the omissions and causal ambiguity of their sources. A useful baseline is often a transparent evidence table that a domain expert can audit. A more complex method earns its place when it finds validated evidence that the table misses, improves prioritisation under a fixed validation budget, or changes a competent analyst's conclusion.

2.3 Assays, endpoints, and entity resolution

Assay data become model-ready only after identity and measurement semantics are resolved. A canonical SMILES string is not a complete experimental identity, because stereochemistry, salt form, tautomer, protonation state, isotopic labelling, mixture composition, and batch can matter. Protein identity may require species, isoform, mutation, construct boundaries, post-translational state, cofactors, and complex partners. An assay endpoint needs protocol, units, relationship type, qualifiers, detection limits, and environmental conditions.

AIDD datasets frequently combine rows that look syntactically compatible but answer different questions. Binding affinity, biochemical inhibition, cell viability, and transcriptional response can all be called "activity". Converting them to a single binary label may increase sample count while destroying interpretability. The defensible order is to retain raw provenance, harmonise only endpoints with an explicit conversion rule, show exclusion counts, and keep enough metadata to reconstruct the decision.

The exclusion counts are part of the result, not a footnote to it. A laboratory that admits 488 measurements from 13,305 reported rows records the 12,817 excluded rows and names the reason for each exclusion class, and a reader can then judge whether the admission rule was reasonable or merely convenient.

2.4 Hit discovery

Hit discovery can begin from physical screening, ligand-based retrieval, structure-based screening, fragment methods, phenotypic screening, repurposing, or generative design. Each route has a different claim. Ligand similarity asks whether neighbours of known actives are useful. Docking proposes geometries and scores under a structural model. A property predictor estimates a learned endpoint within a representation domain. A generator proposes candidates according to an objective and search procedure. None independently proves biochemical activity.

Retrospective virtual-screening results can be inflated by analogue leakage, decoy construction, target redundancy, and knowledge of the benchmark. Docking programs and scoring functions have long shown target-dependent behavior rather than a universal ordering (Warren et al. 2006). A serious evaluation therefore reports the candidate pool, allowed prior information, selection budget, train–test relationship, target state, and prospective confirmation rate. The cheapest useful comparator may be a fingerprint similarity search, a ligand-efficiency rule, a consensus of simple methods, or an experimentally motivated triage filter.

2.5 Hit-to-lead and lead optimisation

Once a chemical series exists, the question changes. The model must support local decisions under sparse, evolving, biased data. Measurements arrive in batches chosen by previous models and chemists, so the training distribution is endogenous. Assays drift, protocols change, and synthesis feasibility constrains which predictions can be tested. Random cross-validation over the accumulated table can be much easier than the next design cycle.

The optimisation target is also vector-valued. Potency, selectivity, solubility, permeability, metabolic stability, clearance, exposure, hERG activity, reactive motifs, synthetic tractability, novelty, and intellectual-property constraints may matter simultaneously. A weighted scalar score can conceal a fatal liability and lets weights be tuned after outcomes are visible. A more legible starting point is a Pareto view with endpoint-specific uncertainty, hard constraints declared in advance, and a record of why each compound was selected.

Active learning and Bayesian optimisation are valuable only when evaluated as policies. A retrospective point-prediction metric does not show that a policy finds better compounds with fewer experiments. The unit of evidence is a design cycle or a realistic replay with a locked acquisition budget. Selection bias, delayed outcomes, failed synthesis, and missing assays must remain visible. Chapter 9 reports a completed replay under exactly these conditions, including the case where the diversity policy lost to every random instance it was compared against.

2.6 Structural and simulation evidence

Structure prediction, co-folding, docking, molecular dynamics, free-energy calculations, quantum chemistry, and machine-learned potentials occupy different rungs of physical approximation. A predicted complex can propose contacts or a binding mode, and it does not establish thermodynamic affinity or cellular effect. Molecular dynamics samples a model under a force field and a finite trajectory. Alchemical calculations can estimate relative or absolute free energies, but their reliability depends on system preparation, sampling, force-field adequacy, mapping, and uncertainty analysis. Community best-practice guidance emphasises these protocol choices (Mey et al. 2020).

The curriculum teaches a ladder of physical and statistical claims, ordered by what each rung can support. A rung that supports a geometric hypothesis does not support an affinity, a score computed under a validated protocol does not transfer to another target, and a free-energy difference computed under a stated model does not establish the accuracy of the model that produced it.

The curriculum therefore teaches a hierarchy of claims. Geometry can support a structural hypothesis. A score can prioritise candidates under a validated protocol. A free-energy estimate can compare a defined perturbation with stated uncertainty. Experimental binding and functional assays are still required to establish the biological effect relevant to the programme.

2.7 Candidate nomination as evidence integration

Candidate nomination is not the point at which every model score is averaged. It is a structured argument that a particular molecule or modality has sufficient efficacy, selectivity, exposure, safety margin, manufacturability, and mechanistic rationale to justify the next investment. Missing evidence must remain missing. Correlated models do not create independent support, and repeated computational seeds do not replace independent biological units.

AIDD Commons represents candidate nomination as a capstone. Learners construct an evidence matrix, identify which claim each experiment supports, expose conflicts, and make a go/no-go recommendation with explicit uncertainty. The aim is not to imitate a company's proprietary

decision template. It is to train disciplined reasoning that remains valid when tools change.

Chapter 3

The AIDD education and infrastructure landscape

3.1 Existing strengths

Open AIDD education already has strong components. TeachOpenCADD organises computer-aided drug-design topics as interactive notebooks using open packages and data, with material spanning acquisition, filtering, screening, structural analysis, machine learning, and uncertainty (Sydow et al. 2019). DeepChem supplies tutorials around a broad molecular-machine-learning library (DeepChem contributors 2026). Therapeutics Data Commons provides task-oriented access to datasets, splits, metrics, and benchmark infrastructure (Therapeutics Data Commons 2026). The European AI-driven Drug Discovery platform and the Open and On-demand Drug Discovery Collaborative add structured education and community routes (AI-driven Drug Discovery 2026; Open and On-demand Drug Discovery Collaborative 2026).

These projects are precedents and dependencies, not competitors to erase. AIDD Commons should link learners to the best existing lesson or tool when it fits. Its distinct job is to connect the full target-to-candidate decision chain, record asset-level openness and evidence, and make evaluation discipline a curriculum spine rather than a late appendix.

3.2 What the registry actually contains

The registry is the widest single artefact in the project, and its shape is worth stating before any claim about it. Figure 5 counts the records by kind. The counts are of records, and a count of records says nothing about whether a record has been verified, which is why the figure carries no status, access, or openness information.

A catalogue of that shape is what a project that teaches decisions, tools, and evidence would hold, and it is not what a project that taught only one layer would hold. It is also a catalogue in which most of the scientific weight sits in records that a reader cannot fully inspect, which is the subject of Chapter 4.

3.3 Gap model

The landscape is better understood by capabilities than by a single “coverage” score.

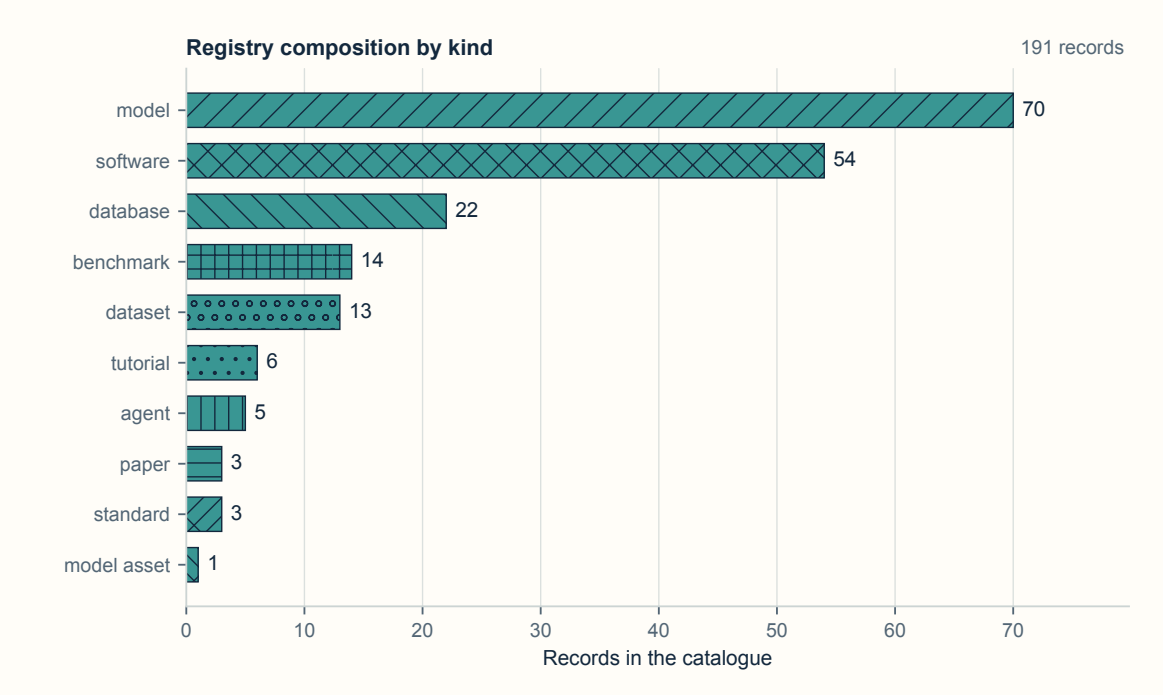


Figure 5: Registry composition by kind. 191 catalogue records: 70 model, 54 software, 22 database, 14 benchmark, 13 dataset, 6 tutorial, 5 agent, 3 paper, 3 standard, 1 model asset. A count of records says nothing about whether a record has been verified, and the figure carries no status, access or openness information.

Capability	Existing open ecosystem	AIDD Commons v0.1 role
Practical CADD notebooks	Strong examples in TeachOpenCADD and tool projects	Link, contextualise, and add protocol-aware bridges
Molecular ML software tutorials	Broad library-specific material	Teach task/data/split invariants across libraries
Dataset and benchmark access	Strong task hubs and individual benchmarks	Add claim-level provenance, applicability, and leakage checks
Biological mechanism to target	Distributed across biology and omics courses	Connect causal evidence to downstream design decisions
Biomolecular frontier models	Fast-moving papers and repositories	Maintain dated, multidimensional openness records
End-to-end curriculum graph	Partial routes across many providers	Publish 18 connected tracks and role-based pathways
Negative results and process knowledge	Inconsistent and often unpublished	Treat experiment reports and corrections as first-class artifacts
English–Chinese entry route	Fragmented	Canonical English plus semantically matched Chinese pathways

This table is a project analysis, not a quantitative audit of every provider. The later comparison audit must define what counts as a topic, exercise, evidence-backed resource, executable lab,

language route, and maintained update before it can support a coverage ranking. Chapter 10 reports that audit and Chapter 11 the reading that checked it.

3.4 The 18-track curriculum

The curriculum covers:

orientation, environments, pathways, and glossary; drug discovery, medicinal chemistry, structural biology, and experimental foundations; machine learning, geometric learning, probability, and scientific computing; data, licensing, standards, cleaning, and entity alignment; molecular, protein, complex, omics, and knowledge-graph representations; disease mechanism, target discovery, causal inference, and perturbational omics; QSAR, activity prediction, and molecular properties; ADMET, toxicity, DMPK, and PK/PD; drug–target interaction, affinity, virtual screening, and repurposing; structure prediction, pockets, and co-folding; docking, scoring, molecular dynamics, free energy, quantum chemistry, and ML potentials; small-molecule generation, search, and multi-objective optimisation; protein, peptide, antibody, nucleic-acid, and multimodal design; reaction prediction, retrosynthesis, synthesizability, and laboratory automation; active learning, Bayesian optimisation, and design–make–test–analyse loops; scientific language models, retrieval, tools, and agents; splits, benchmarks, uncertainty, out-of-distribution evaluation, leakage, and reproducibility; and candidate nomination, failure analysis, biosecurity, open science, and capstone work.

Figure 6 counts the learning units in each track by layer and sets the lesson count beside it. A unit is a learning objective attached to a lesson, so the two counts differ, and neither says how long a track takes or whether a learner reaches its end.

Tracks are ordered but not isolated. Data and evaluation concepts recur because a learner needs them before any domain-specific result can be trusted. Biological foundations connect to target discovery, chemical foundations connect to property prediction and synthesis, and structural foundations connect to co-folding, docking, dynamics, and design. The final track requires learners to integrate these branches rather than treating completion of notebooks as competence.

The curriculum registry records 120 learning units across 100 lesson nodes, in 56 core, 44 research, and 20 practice layers. That is a count of what is registered, not a measure of quality.

The two columns move against each other, and that is the useful part of the table. Track 11 carries eight lesson nodes against eight learning units, while track 13 carries nine lesson nodes against eight units. A track with many lessons and few units is a track whose lessons carry broad objectives, and a track with many units and few lessons is a track whose material is decomposed finely. Neither is better, and the reader who wants to know what a track costs in time has to look at the lessons themselves.

3.5 Pathways by reader

Biology-first readers begin with orientation, chemical and structural foundations, scientific computing, and data contracts. They then follow disease mechanism and target discovery before choosing property, interaction, or modality-specific design tracks. Their common failure mode is accepting a computational representation without checking what biological distinctions it removes.

AI-first readers begin with the drug-discovery decision map, experimental foundations, data licensing and entity alignment, then representations and evaluation. They should encounter scaffold, target, time, and subject splits before complex architectures. Their common failure mode is treating rows as independent samples and metric gain as scientific utility.

Medicinal chemists and structural scientists can enter through the target-to-candidate

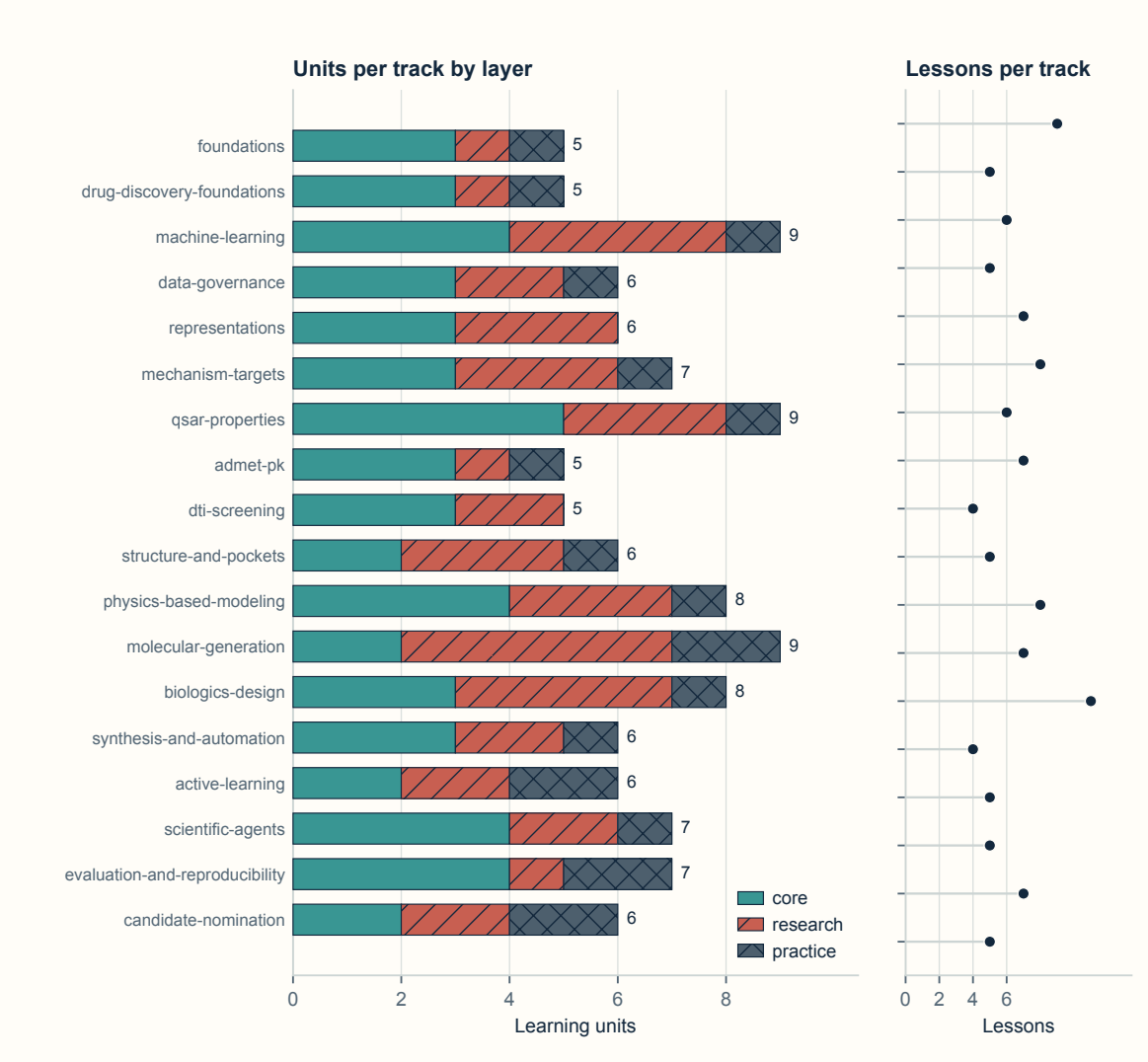


Figure 6: Curriculum units per track across the 18 tracks, split by layer, beside the number of lessons in each track. The curriculum holds 120 learning units in 114 lessons, 56 core, 44 research and 20 practice. A unit is a learning objective attached to a lesson, so the two counts differ, and neither one says how long a track takes or whether a learner reaches its end.

map, data semantics, property and interaction prediction, structure and simulation, and multi-objective optimisation. Their common failure mode is using ML uncertainty as if it captured assay, domain, and model-form uncertainty together.

Engineers and research-software developers begin with environment and data contracts, then registry, reproducibility, model-serving, language-model tools, and safety. They should still complete one biological decision slice and one honest-baseline lab. Their common failure mode is building reliable infrastructure around an invalid scientific target.

The prerequisite relations the curriculum registry states are drawn in Chapter 5, where the ordering of a learner's route becomes a question about data contracts.

3.6 Infrastructure as curriculum

The platform itself teaches. Each Atlas entry is one reviewable YAML record. Generated JSON and CSV expose deterministic transforms. Query parameters preserve Atlas filters. Learning progress remains local to the browser. Labs publish exact environments and commands. Public releases are built from an allowlist so internal operational material is not accidentally shipped.

These choices make hidden assumptions inspectable. A learner can see why a resource is marked `unverified`, how a relation resolves, and which source supports a licence. An engineer can reproduce the catalog export without needing a database service. A reviewer can distinguish canonical content from a generated surface. The architecture is intentionally modest so that scientific curation, not platform maintenance, remains the dominant work.

The depth distribution of the curriculum is not uniform, and Figure 13 in Chapter 6 is the figure a reader should use to decide where to look next.

The run basis is what changes what a reader can do. Forty-three of the fifty-six runnable topics rest on measured or computed records that exist independently of the course, and thirteen rest on hand-written or synthetic inputs, so a reader who wants to practise on real chemistry has forty-three topics to choose from and a reader who wants to practise the mechanics of a decision on constructed inputs has thirteen. Neither number says whether a learner succeeded, because depth records what a reviewer observed in the material.

Chapter 4

The open-development frontier

4.1 From open artifacts to open process

An open repository is valuable, but it is not a complete account of how a frontier system was built. Open Athena describes *process knowledge* as the accumulated choices about data, experiments, optimisation, systems, evaluation, failed routes, and scale-up that rarely enter a final paper or model release (Hall 2026). Marin operationalises a version of this idea by making code, weights, data sources, experiments, failures, and tools visible as the work proceeds (Marin contributors 2026).

AIDD has the same structural problem, amplified by experimental and commercial constraints. A model may be accompanied by inference code and weights while its training data, negative experiments, selection logic, data cleaning, benchmark exclusions, or laboratory feedback remain private. A closed programme may disclose a scientific architecture but not its operational system. Academic work may publish code only after acceptance and omit failed screens or unstable recipes. Regulatory, patient, partner, and licence constraints can also make full disclosure impossible for legitimate reasons.

The goal is therefore not a binary moral label. It is a legible account of which assets and decisions are available, under what terms, and what remains unknown.

4.2 Seven-dimensional openness

AIDD Commons records seven dimensions separately:

Inference code, which asks whether a user can execute the released model or method; **training code**, which asks whether a user can train or fine-tune through the claimed procedure; **weights**, which asks whether the parameters are obtainable and under which terms; **architecture**, which asks whether the computational design is described sufficiently for scrutiny or implementation; **data recipe**, which asks whether sourcing, filtering, preprocessing, mixtures, and exclusions are documented; **training data**, which asks whether the actual training examples or a reconstructable lawful equivalent are available; and **evaluation code**, which asks whether the reported evaluation can be reproduced under the stated protocol.

Lifecycle and access are separate. A resource can be active and closed, legacy and open, gated with source available, or public only through an API, and each non-`unverified` value needs direct evidence. The Open Source AI Definition provides a broader normative account of freedoms and preferred forms for modification (Open Source Initiative 2024), and the Atlas dimensions serve a narrower empirical purpose by showing which AIDD assets were actually evidenced in a dated review.

`Available` is not synonymous with unrestricted, since code, weights, data, and documenta-

tion can carry different licences. `Not available` requires evidence that the asset is withheld or absent within a defined scope, and when a reviewed source says nothing the correct state is `unverified`.

Figure 7 draws the model of the seven dimensions beside two catalogue records drawn as the forms they are, a model and a dataset. They are close to opposites. Where the model has inference code and weights, the dataset has nothing to open there, and the dataset's one available dimension is the data recipe. The figure also carries the vocabulary of the four states and the two axes the schema holds separate from openness.



Figure 7: Seven-dimensional openness. (a) and (b) are two catalogue records drawn as the forms they are, a model and a dataset, and they are close to opposites. Where the model has inference code and weights, the dataset has nothing to open there, and the dataset's one available dimension is the data recipe. (c) is the vocabulary of the four states, available, not available, not applicable and unverified, where unverified records that no claim is made. (d) is the two axes the schema holds separate from openness, lifecycle status and access level. The two records are a dated snapshot from the catalogue review of 21 October 2026, and a later re-review can change their states.

Figure 8 gives the same seven dimensions across all 191 records, grouped by kind. The fill is the share of a kind's records for which a dimension is available, counted over the records for which it applies, because a dimension that does not apply to a record kind is not evidence about it. The hatch says why the rest is not available.

The largest cell in that matrix is unverified, and that is a statement about what has been read rather than about what exists. A sweep on 23 October 2026 read the file listing and the README of every repository whose seven dimensions were all unverified, and the cells that changed are the ones a reader can check by opening the repository: the inference code of 65 of 70 models, the training code of 41, the architecture of 46 and the evaluation code of 15, where before the sweep those cells stood at 40, 25, 31 and 6. A second sweep on the same day read the training-data statement of every model page that carries one, and the data recipe rose from 6 of 70 records to 17, with the training data dimension at 20. The data recipe rose from 6 of 70 records to 21

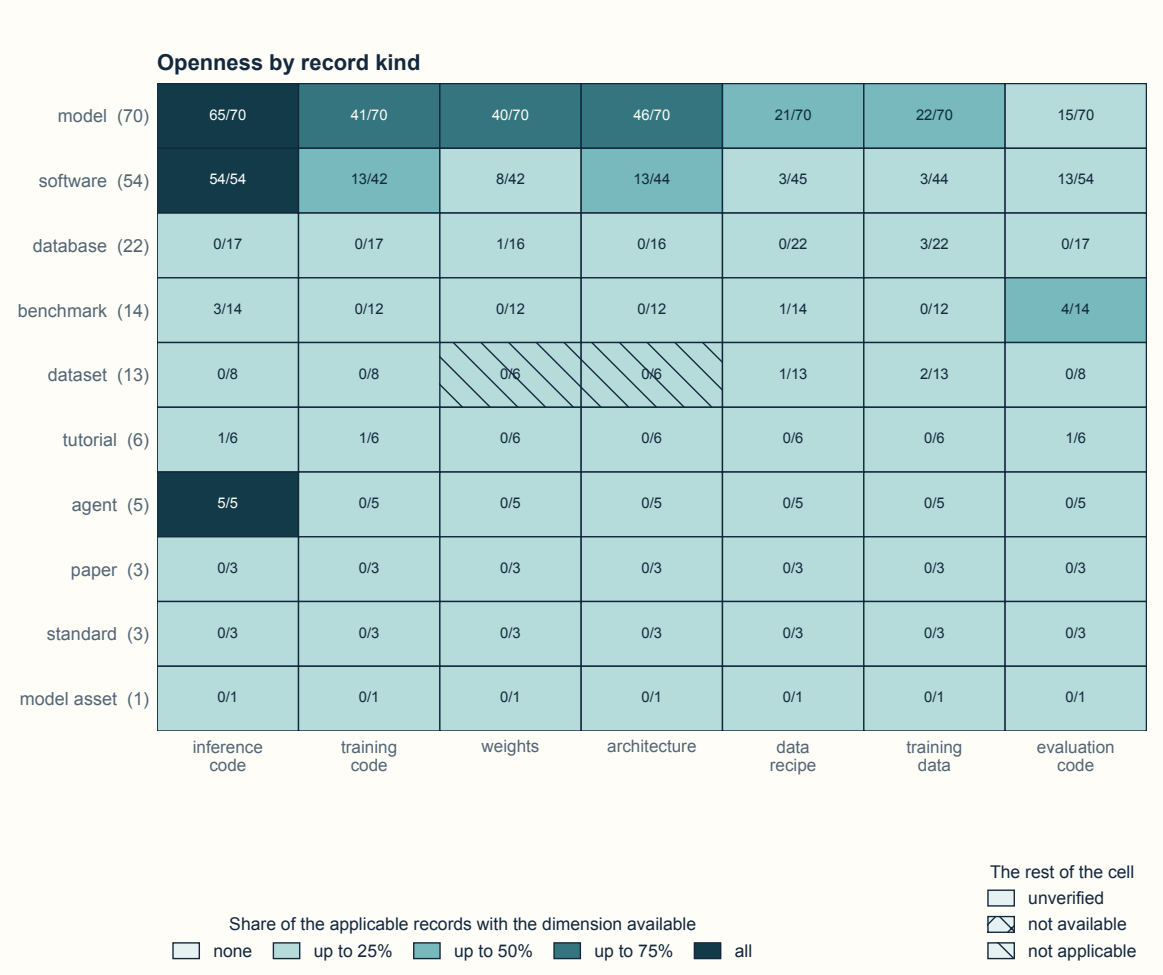


Figure 8: Openness by record kind and by the seven openness dimensions, over 191 catalogue records. The fill is the share of a kind's records for which a dimension is available, counted over the records for which it applies, and the hatch says why the rest is not available, unverified, recorded as not available, or not applicable to that kind. Openness is recorded per dimension and is never collapsed here into a single open or closed label.

and the training data dimension from 18 to 22. The data recipe is still the weakest dimension for models, and it is still the most consequential gap for a reader who wants to know what a model was trained on, because most repositories name a corpus without stating the procedure that built the training set. And the two dimensions that are hardest to establish are the two a reader most needs, which is the argument for recording them separately rather than collapsing them.

Table 3: Openness by record kind, counted over the 191 catalogue records.

Kind	Records	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code
Model	70	65	41	40	46	6	18	15
Software	54	54	13	8	13	3	3	13
Database	22	0	0	1	0	0	3	0
Benchmark	14	3	0	0	0	1	0	4
Dataset	13	0	0	0	0	1	2	0
Tutorial	6	1	1	0	0	0	0	1
Agent	5	5	0	0	0	0	0	0
Paper	3	0	0	0	0	0	0	0
Standard	3	0	0	0	0	0	0	0
Model asset	1	0	0	0	0	0	0	0

4.3 Access levels and licence layers

Openness is one axis. Access level and licence are two more, and they are recorded independently. Figure 9 counts records by access level and by licence layer.

The two numbers a reader needs in order to size the remaining work are the records that carry no verified access level and the records that carry no licence. A sweep on 23 October 2026 opened the terms page of every record the first sweep had left unverified and cut the first number from 110 to 46 and the second from 104 to 34, so the two bars are now smaller than the open-source and permissive-licence bars rather than larger, and the remaining debt is the smaller half of the figure.

Table 4: Access level and licence layer across the 191 catalogue records.

Access level	Records	Licence layer	Records
Open source	125	Permissive software licence	101
Source available	15	Copyleft	22
Research only	4	Creative Commons attribution	18
Gated	3	Creative Commons non-commercial	6
Open weights	1	Other stated terms	8
Closed	1	Public domain dedication	5
No verified access level	42	No licence recorded	31

Twenty-two records carry more than one licence, and they are counted once each in the most restrictive layer they carry. The 31 records with no licence recorded are not records with no licence. They are records whose licence has not been read, which is why the bar is drawn empty rather than coloured, and a reader who wants to redistribute one of them has to open its own terms.

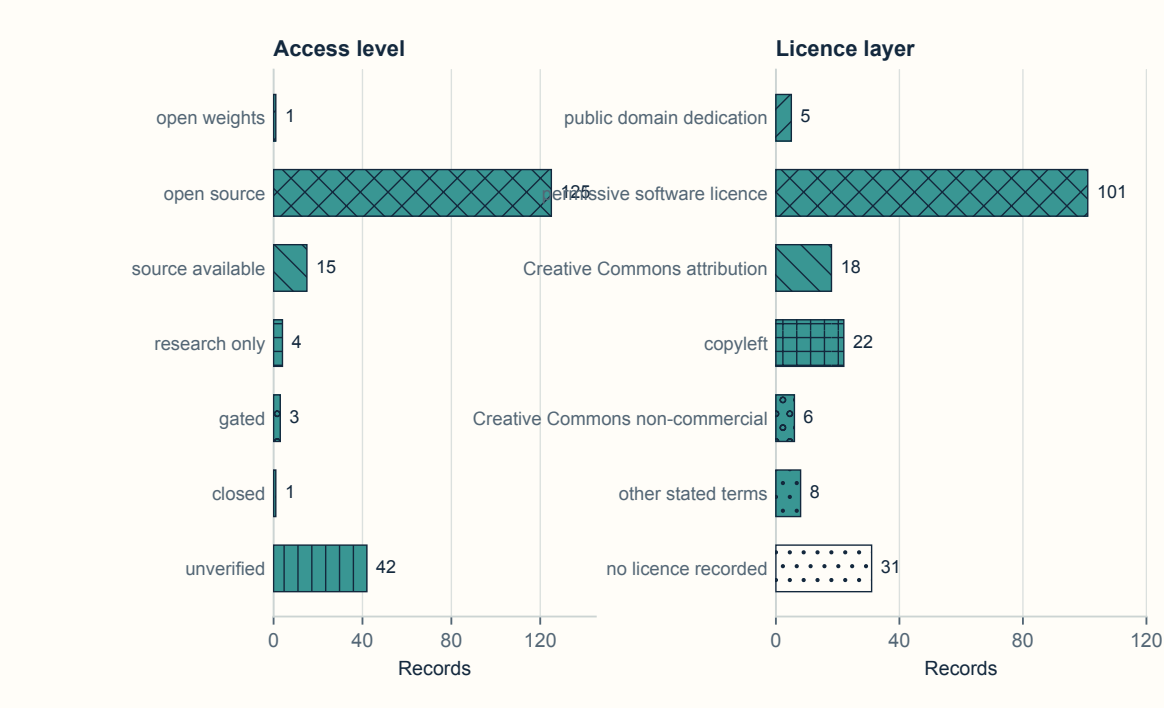


Figure 9: Access level and licence layer for the 191 catalogue records. 42 records carry no verified access level and 31 record no licence. The verified access levels are 1 open weights, 125 open source, 15 source available, 4 research only, 3 gated, 1 closed. By licence layer, 5 public domain dedication, 101 permissive software licence, 18 creative commons attribution, 22 copyleft, 6 creative commons non-commercial, 8 other stated terms. A record with more than one licence is counted once, in the most restrictive layer it carries.

4.4 Co-folding as a frontier case

Biomolecular co-folding exposes why a binary open/closed label fails. AlphaFold established a major change in protein-structure prediction (Jumper et al. 2021), and AlphaFold 3 extended structure prediction across biomolecular interactions (Abramson et al. 2024). Subsequent systems span academic, industrial, open-source, gated, and preview release patterns. Their tasks are also not identical, since some focus on complex structure, some add affinity prediction (Passaro et al. 2025), and some position themselves as broader design engines (Chai Discovery et al. 2024).

The v0.1 Atlas reviewed the following official surfaces on 4 September 2026. The statements below describe evidenced availability, not scientific performance.

AlphaFold 3 inference pipeline (Google DeepMind 2026) provides public inference code under an Apache-2.0 code licence, and its README states that the model parameters are obtained separately under their own terms, so the record does not infer training-code, training-data, or full-recipe availability from the paper. **Chai-1** (Chai Discovery 2026b) provides inference material in the official `chai-lab` repository and describes downloadable weights, which supports code-and-weight availability while leaving a complete training recipe and training data unverified. **Chai-2** (Chai Discovery 2026a) describes limited academic early access on its product page, which establishes neither public code, weights, training data, nor asset licences, so those dimensions remain unverified rather than being guessed from the product identity. **IsoDDE** (Isomorphic Labs 2026) is described by a company announcement as a proprietary AI and data engine across discovery stages, with no asset-level evidence for code, weights, recipes, or data, so the record states closed access at the described service level and leaves the individual dimensions unverified. **Boltz-1 and Boltz-2** (Boltz contributors 2026) provide inference code and weights under the terms recorded in the common repository, and for the reviewed snapshot complete training and updated evaluation materials were not established. **OpenFold** (OpenFold contributors 2026) presents an open AlphaFold-style reproduction with code and model assets, and the record supports public inference and weights without upgrading every training-data and recipe dimension. **Protenix** (Protenix contributors 2026) describes both training and inference material and documents data-related components, and the record preserves an asset-level licence caveat for exact data subsets. **OpenDDE Preview** (Aureka Research 2026) describes a preview all-atom engine with source and downloadable weights and warns that interfaces and checkpoints may change, so the training process and data dimensions remain unverified.

The scope of the licence that was read is the field a reader who intends to reuse an asset has to check, and it differs from the licence name in most of these records.

Table 5: The co-folding frontier as the registry records it, after the review of 4 September 2026.

Record	Access level	Dimensions the record establishes	Scope of the licence that was read
AlphaFold 3 inference pipeline	Source available	Inference code	Apache-2.0 over the code, not over the parameters
Chai-1	Open source	Inference code, weights, architecture	Apache-2.0 over the source and the weights
Chai-2	Gated	None	No asset licence established
IsoDDE	Closed	None	No asset licence established

Record	Access level	Dimensions the record establishes	Scope of the licence that was read
Boltz-1 and Boltz-2	Open source	Inference code, weights; training code for Boltz-1	MIT over the source and the weights, not over the preprocessed structures
OpenFold	Open source	Inference code, weights	Apache-2.0 over the source, CC BY 4.0 over the parameters
Protenix	Open source	Inference code, training code, weights, architecture, data recipe, training data	Apache-2.0 over the source and the v1 parameters, proprietary terms over the v2 parameters
OpenDDE Preview	Open source	Inference code, weights	Apache-2.0 over the source, checkpoints under their own terms

This snapshot deliberately avoids a performance ordering. Published numbers may use different datasets, dates, templates, inference budgets, ligand information, target classes, leakage controls, and post-processing. A scientifically meaningful comparison would require a matched evaluation suite, frozen inputs, common exclusions, identical structural templates and databases where possible, and uncertainty over independent complexes or targets. The registry has no field for such a comparison, because the registry records assets rather than performances.

4.5 What should be disclosed

For a computational AIDD result to become reusable process knowledge, a release should aim to disclose:

the decision and hypothesis rather than only the final model; exact data sources, versions, filtering, entity normalisation, and exclusions; split-generation code and leakage checks; baselines with recipe parity; training and inference commands, environments, seeds, and resource class; metric definitions and aggregation units; model-selection and stopping rules fixed before test access; failed experiments and why they were not advanced; calibration, domain-of-applicability, and prospective validation plans; and separate terms for code, weights, data, and derived artifacts.

Not every item can always be public. When patient privacy, commercial agreements, controlled access, safety, or licences constrain disclosure, a useful report can still describe the missing asset, its effect on reproducibility, and a public proxy that does not overstate equivalence.

Each item on that list unblocks a specific reader decision. A decision and hypothesis say what question the artefact answers, and data sources and exclusions say what it does not contain. Split code and leakage checks say whether the evaluation measures transfer or recall, and baselines with recipe parity say whether the method beats the cheapest serious alternative. Commands and seeds say whether the run can be repeated and what it costs. Selection rules fixed before test access say whether the result was chosen after seeing the answer, failed experiments say whether a route was already abandoned, and separate terms for code, weights, and data say whether the reader may use each asset for the intended purpose.

4.6 Process knowledge as a community artifact

The contribution system treats experiment reports, corrections, and RFCs as first-class work. A negative result is useful when the hypothesis, protocol, and failure are legible. A correction is useful when it propagates to every surface that repeated the claim. An RFC is useful when it preserves alternatives and dissent rather than presenting a decision as inevitable.

This changes the meaning of "tutorial". The reader does not only reproduce a successful notebook. They see how the task was chosen, why a baseline was retained, what leakage check could fail, when the test set was opened, what the observed output was, and which conclusion the evidence does not support. Chapter 9 is written to that standard, and every laboratory it reports states what the run does not show.

Chapter 5

Data, standards, tasks, and benchmarks

5.1 The data contract precedes the model

Every modeling task should begin with a compact contract:

Field	Question
Decision	What action could change because of the output?
Entity	What exactly is one molecule, protein, complex, cell, patient, or reaction?
Input	Which measurements and metadata are available at prediction time?
Label	How was the target measured or constructed, with what units and censoring?
Independent unit	What unit supports uncertainty and generalisation claims?
Split	Which novelty or deployment boundary is simulated?
Baseline	What is the strongest simple method using the same allowed information?
Metric	Which error or utility corresponds to the decision?
Missingness	Why are values absent, and can that absence leak the answer?
Licence	Can the data be accessed, transformed, redistributed, and used as proposed?

FAIR principles make data findability, accessibility, interoperability, and reusability explicit (Wilkinson et al. 2016). In AIDD, however, FAIR metadata alone does not guarantee fitness for a specific prediction. A dataset may be easy to download but merge incompatible assays, omit negative measurements, confound target and chemical series, or forbid redistribution of derived records.

The contract is a precondition for a fair comparison, and the last two fields are the ones a reader of a published result most often has to reconstruct for themselves. A missingness rule that can leak the answer is the standard way a curated dataset turns into a benchmark that measures curation rather than chemistry, and a licence that forbids redistribution of derived records can prevent a reader from checking the entity resolution that the contract declares.

5.2 Identity layers

Chemical identity should preserve the original record alongside standardised forms. Canonicalisation choices include charge, salts, solvents, tautomer handling, stereochemistry, isotopes, mixtures, and duplicate resolution. The identifier used for deduplication may differ from the one used for learning. A graph model cannot recover stereochemistry that preprocessing removed.

Biological identity needs equally careful layers, from gene identifiers through transcript and protein accessions, isoforms, species, mutations, constructs, post-translational modifications, complexes, conformational states, tissues, cell lines, donor context, and disease ontology. Mapping everything to a gene symbol can make joins convenient while silently merging different intervention targets.

Reaction identity includes reactants, reagents, solvents, catalysts, products, stoichiometry, temperature, time, yield definition, failed reactions, and provenance. The Open Reaction Database illustrates an open schema and repository for structured reaction data (Kearnes et al. 2021). Even with a schema, absence of unsuccessful reactions or procedure details can limit what a synthesis model learns.

Each identity layer has a specific loss that a canonicalisation can cause. A chemical record that keeps charge, salt, tautomer, stereochemistry, isotope, mixture, and batch can lose the distinction a flat representation no longer carries. A biological record that keeps gene, transcript, protein, isoform, species, construct, modification, complex, state, tissue, donor, and disease ontology can lose two intervention targets merged into one label. A reaction record that keeps reagents, catalysts, conditions, yield definition, failures, and provenance can lose the evidence that a synthesis model learns only where synthesis succeeded.

5.3 Task families

The Atlas groups resources by task while allowing multiple pipeline stages and modalities.

Property and activity prediction includes physicochemical properties, QSAR, bioactivity, solubility, permeability, metabolism, toxicity, and clearance. Evaluation must preserve units, censoring, assay identity, and chemical-series boundaries.

Interaction and structure includes drug–target interaction, affinity, pocket detection, docking, pose prediction, co-folding, dynamics, and free-energy estimation. Targets, structures, templates, ligand conformers, and structural databases can all leak related examples.

Generation and optimisation includes de novo molecule design, scaffold decoration, protein and peptide design, and multi-objective search. Validity and novelty are necessary diagnostics but not substitutes for property, synthesis, or experimental evaluation.

Synthesis and automation includes reaction prediction, retrosynthesis, condition recommendation, yield prediction, procedure extraction, robotics, and closed-loop optimisation. Evaluation should distinguish exact dataset reconstruction from practical route quality and execution success.

Biology and translation includes target prioritisation, perturbation response, omics integration, causal inference, and biomarker work. The independent unit often lies above the row or cell level, so donor, cohort, laboratory, and time separation are load-bearing.

Language models and agents include retrieval, literature synthesis, tool use, workflow planning, and autonomous experimentation. Their evaluation must separate factuality, tool execution, scientific decision quality, and end-to-end outcome. A fluent explanation is not evidence that the cited source was opened or that an action was correct.

Each family has an independent unit its evaluation usually rests on, and a leakage route that most often decides the result. Property prediction rests on the compound within a declared series

and leaks through near-analogues across a scaffold split. Interaction and structure rests on the target or complex and leaks through templates and related targets. Generation and optimisation rests on the sampled candidate and leaks through oracle exploitation. Synthesis rests on the route or reaction and leaks through reagent overlap. Biology and translation rests on the donor, cohort, or laboratory and leaks through cell-level pooling of donor replicates. Language models and agents rests on the task and its grading rule and leaks through contamination and tool-state dependence.

5.4 Benchmark choice

Benchmarks are valuable because they stabilise data, splits, metrics, and baselines. MoleculeNet helped standardise evaluation across molecular machine-learning datasets (Wu et al. 2018), and TDC organises therapeutic tasks and benchmark interfaces (Therapeutics Data Commons 2026; Huang et al. 2021). But a benchmark's protocol defines the claim. Replacing its split, removing difficult rows, changing target aggregation, or tuning on the test set creates a different task even when the dataset name is unchanged.

Before adopting a benchmark, ask:

Does its endpoint match the scientific decision; are train and test inputs available under the intended deployment conditions; is the independent unit represented by the split; are duplicates, close analogues, targets, structures, patients, and time handled appropriately; is the strongest reproducible baseline included with recipe parity; are missing labels and exclusions visible; and are data and code terms compatible with teaching and redistribution?

Ligand-based benchmarks can reward recognition of chemical similarity rather than generalisation to meaningfully new chemistry (Wallach and Heifets 2018). A time split can better imitate prospective use in some medicinal-chemistry settings, though it also changes assay mix and programme context (Sheridan 2013). Scaffold splits target a different notion of novelty and can be unstable for small datasets. No split is universally "realistic", and it must be connected to a deployment claim.

The seventh question has an additional consequence for a teaching project. A benchmark whose terms forbid redistribution cannot be carried in a public release, which means a lab built on it cannot be handed to a reader as a runnable exercise. The three licensed clinical benchmark suites this project registers, ChemBench, LAB-Bench, and AI-Researcher, differ in exactly this respect, and Chapter 12 reports what each of their records establishes about reuse (Mirza et al. 2024; Laurent et al. 2024).

5.5 Baselines and representations

Cheap baselines are diagnostic instruments. For molecular regression they may include the training mean, physicochemical descriptors with linear or tree models, and circular fingerprints with regularised linear models or random forests. Extended-connectivity fingerprints encode local neighbourhoods and remain a strong reference representation (Rogers and Hahn 2010). For sequence tasks, k-mer, profile, or pretrained embedding baselines can expose whether a new architecture contributes beyond representation scale. For knowledge graphs, degree, frequency, and path heuristics can reveal shortcut structure. For omics, cell-type means and linear perturbation effects often deserve comparison before a large generative model.

The strongest cheap baseline is not necessarily the simplest individual model. If two allowed signals are complementary, a train-only blend or small ensemble is a legitimate baseline. Hyperparameter effort, preprocessing, external information, and inference budget should be comparable before an architectural gain is interpreted.

A learned encoder is judged on the same grounds as anything else, and the comparison that decides it is against a representation of comparable or greater capacity. The grid-transformer family made a graph attention mechanism a standard component rather than a distinguishing one (Ying et al. 2021), which is the reason the curriculum teaches representations and their baselines together rather than in sequence.

5.6 A minimum benchmark card

Every benchmark-facing lesson should publish:

task and intended use; source versions and acquisition dates; row and independent-unit counts before and after filtering; entity, endpoint, units, missingness, and censoring rules; split generator, grouping fields, and overlap audits; allowed external data and pretrained models; baseline recipes and model-selection budget; metric implementation and aggregation; uncertainty method and independent resampling unit; known leakage routes and domain limitations; and asset-specific licences and redistribution boundary.

This card is deliberately stricter than a table of dataset names. It lets a reader decide whether two reported numbers answer the same question.

5.7 Prerequisites as a data dependency

The curriculum is a graph rather than a sequence, and the graph is drawn from the same evidence standard as everything else. Figure 10 draws the prerequisite relations the curriculum registry states, over the six learning paths. Ten distinct lessons are named as a prerequisite somewhere in the catalogue, and twenty-one prerequisite relations connect them to the five specialist routes. A solid edge is a prerequisite the catalogue states, and a dashed edge is a lesson the core route also lists, which is a different relation and is drawn as a different relation. The core route names no prerequisite of its own.

A reader who looks at the badges rather than the edges sees which lessons are hubs. A lesson required by four of the six routes carries more of the curriculum's weight than a lesson required by one, and a hub that teaches a data contract is the one whose absence would break the most routes. The figure counts routes, not difficulty, and it does not say how long a route takes.

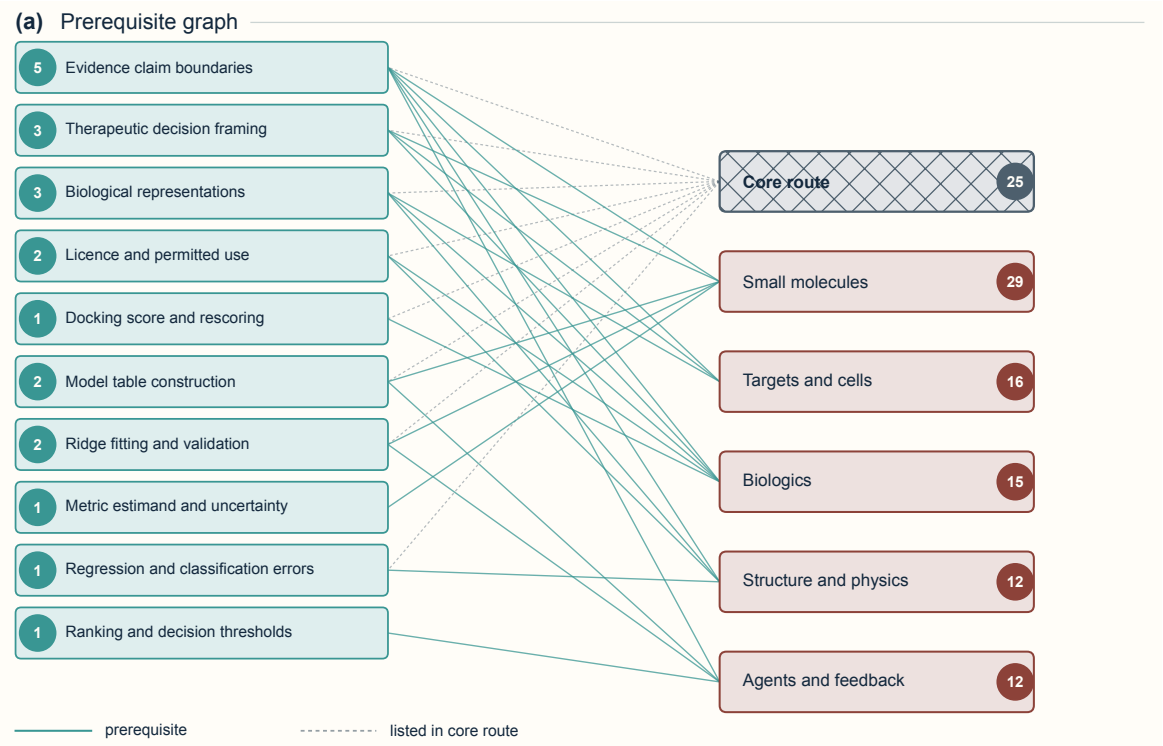


Figure 10: Prerequisite graph over the six learning paths. Ten distinct lessons are named as a prerequisite somewhere in the catalogue, and twenty-one prerequisite relations connect them to the five specialist routes. A lesson's badge gives the number of routes that require it, so the hub is visible without reading the edges, and a route's badge gives the number of lessons the route lists. A solid edge is a prerequisite the catalogue states. A dashed edge is a lesson the core route also lists, which is a different relation and is drawn as a different relation. The core route names no prerequisite of its own. The badges are counts from the catalogue, not difficulty, depth or coverage.

Chapter 6

Reproducibility, evaluation, and failure modes

6.1 Reproducibility is claim-specific

Reproducibility has several layers. A command may reproduce a file. A notebook may reproduce a metric. An independent implementation may reproduce a method. A new dataset or prospective experiment may reproduce a scientific conclusion. These are increasingly demanding claims and should not be collapsed into a single badge.

A computational artifact should expose the environment, exact command, deterministic or stochastic components, data acquisition, preprocessing, split, fitting scope, outputs, and expected checks. It should also show where determinism ends. GPU kernels, external APIs, mutable databases, stochastic conformer generation, and upstream model changes can create variation even when a random seed is fixed.

6.2 Frozen evaluation

The central rule is to freeze the protocol before looking at the held-out test result. The freeze includes task definition, eligible samples, labels, split, preprocessing, baseline recipes, hyperparameter selection, metrics, calibration, seed aggregation, and stopping rule. Training and validation data can be used as declared. The test set is opened once to estimate the frozen procedure's performance.

This rule prevents several common forms of silent overfitting, including choosing a seed after test inspection, adding exclusions that improve the result, changing the primary metric, tuning conformal coverage, selecting the best checkpoint on test, and presenting the friendliest of several splits. If a post-test bug is found, the report should distinguish a protocol-invalidating error from a new scientific hypothesis and record why another evaluation is justified.

Figure 14 draws the frozen lifecycle as a sequence, from the written protocol through the freeze, the structural and finite-output pretests on training and validation data, the single opening, to the reviewed summary. A bug found after the test has been opened is reported as a protocol-invalidating error, and the barred loop is the prohibition on a second opening. The figure carries no measured value, and the two pretests are drawn as they pass rather than as they can fail. It is reproduced in Chapter 9, where the eight protocols that follow it are read one by one.

6.3 FreeSolv honest-baseline laboratory

The first executable lab uses FreeSolv version 0.52, a database of experimental and calculated hydration free energies (Mobley and Guthrie 2014; MobleyLab 2017). The lab is intentionally modest. Its purpose is to expose split effects, train-only fitting, baseline strength, and conformal interval behaviour on a CPU-feasible molecular regression task.

Project decision. The frozen dataset contains 642 parsed molecular records. Both random and Bemis–Murcko scaffold protocols use 514 training, 64 validation, and 64 test records. Scaffold grouping follows the molecular-framework concept introduced by Bemis and Murcko (Bemis and Murcko 1996). The baselines are the training mean, Morgan fingerprints with ridge regression, and Morgan fingerprints with a random forest. Conformal radii are fitted only on validation residuals. The primary split is scaffold, and MAE and RMSE are primary point metrics.

Observed result. After structural checks and validation runs passed, the held-out test evaluation was executed once. The following values are the frozen output, and they were not selected from repeated test runs.

Split	Model	MAE (kcal/mol)	RMSE (kcal/mol)	Coverage	Mean interval width
Scaffold	Training mean	2.591	3.288	—	—
Scaffold	Morgan + ridge	1.128	1.760	0.875	5.479
Scaffold	Morgan + random forest	1.257	2.038	0.906	6.013
Random	Training mean	2.767	3.450	—	—
Random	Morgan + ridge	1.019	1.351	0.734	3.165
Random	Morgan + random forest	1.368	1.940	0.875	5.158

The scaffold split has no overlap in its grouping keys across train, validation, and test. The random split has no molecule-identity overlap, but related scaffold groups can occur across partitions by design. The conformal radii were 2.739 kcal/mol for scaffold ridge, 3.007 for scaffold random forest, 1.583 for random ridge, and 2.579 for random forest.

These results support limited conclusions. Both fingerprint models improve point error over the training mean on both frozen splits. Ridge has lower MAE and RMSE than the random forest in this run. The random ridge interval coverage of 0.734 is below its nominal target in the finite held-out sample, and the lab reports it rather than recalibrating after test access. None of the results establishes state of the art, applicability to other endpoints, or prospective medicinal-chemistry utility.

The split comparison is the point of the exercise. The same model, the same features, and the same fitting code produce 1.128 kcal/mol on the scaffold split and 1.019 kcal/mol on the random split. A number quoted without its split does not identify an experiment. The figure that separates the point metrics from the interval metrics is in Chapter 9, where the same six rows are read alongside the other seven protocols.

6.4 Uncertainty and calibration

A point error averaged over molecules does not tell a chemist whether an individual prediction is reliable. Uncertainty methods should state what randomness or error source they represent. Deep ensembles, Bayesian approximations, quantile models, and conformal methods answer different questions and can all fail under distribution shift.

For split conformal regression, calibration residuals define an interval under exchangeability assumptions and a specified finite-sample rule. Coverage on a small test set can depart from the target. Reporting only mean width can also hide subgroup failure. A serious analysis inspects coverage and error by chemical region, target class, assay, or other deployment-relevant group without using the test findings to rewrite the frozen headline protocol.

The FreeSolv intervals make the point in miniature. The scaffold ridge reaches a coverage of 0.875 at a mean width of 5.479 kcal/mol, and the random ridge reaches 0.734 at a width of 3.165. The narrower interval is also the less accurate one, which is the ordinary behaviour of a method whose calibration was fitted on a validation distribution that does not match the test distribution.

6.5 Failure-mode taxonomy

Identity failure. Duplicate compounds, alternate salts, protein aliases, overlapping patients, or derived samples cross a split. The cheapest check compares canonical entity and grouping keys across partitions.

Endpoint failure. Labels with different physical meaning or units are pooled. The cheapest check stratifies provenance, units, assay type, and qualifier before modelling.

Temporal failure. Future information enters features, curation, or labels. The cheapest check reconstructs when every input became available and applies a time-aware split.

Representation shortcut. A model exploits target, scaffold, batch, source, or missingness patterns instead of the intended mechanism. The cheapest check is a simple shortcut baseline and counterfactual masking or grouping analysis.

Protocol mismatch. A claimed baseline uses different preprocessing, external data, tuning budget, or evaluation code. The cheapest check is recipe parity on the same implementation path.

Selection leakage. Test performance changes model choice, threshold, metric, or data cleaning. The cheapest check is a dated protocol and a single controlled evaluation entry point.

Uncertainty failure. Repeated seeds are treated as independent biological samples, or intervals are calibrated on test data. The cheapest check names the independent unit and traces every calibration input.

Reproducibility failure. A notebook works only with hidden files, a mutable URL, or a local environment. The cheapest check runs it in a clean CPU environment from documented acquisition instructions.

Scientific overreach. Retrospective prediction is presented as mechanism, binding, efficacy, or candidate quality. The cheapest check writes the next independent experiment that could falsify the claim.

Each failure mode has a cheapest check that detects it, and each check corresponds to an artefact in this repository that shows it ran. The identity check is the split membership files, and the decision-unit lab counts 71 salt-form measurements among 488 admitted points. The endpoint check is the admission funnel with its exclusion counts. The temporal check is a dated protocol and acquisition record, and the L1000 lab fixes one dose and one timepoint out of the series it reads. The shortcut check is a baseline that loses to its own control, and the FreeSolv forest loses

to the ridge on the primary split. The parity check is one protocol naming every baseline, and the ChEMBL EGFR lab fits three baselines through one path. The selection check is a locked summary with a single open, and every reviewed summary in Chapter 9 states its one opening. The unit check is the correlated-frame standard errors, and the rMD17 page reports a block-averaged standard error 3.30 to 6.26 times the naive one. The overreach check is a stated scope in the summary, and every lab in Chapter 9 states what it does not show.

Table 6: The failure modes, their cheapest checks, and where this repository has met them.

Failure mode	Cheapest check
Identity failure	Compare grouping keys across partitions
Endpoint failure	Stratify provenance, units, and qualifiers
Temporal failure	Reconstruct when every input became available
Representation shortcut	A shortcut baseline and grouping analysis
Protocol mismatch	Recipe parity on one implementation path
Selection leakage	A dated protocol and one evaluation entry point
Uncertainty failure	Name the independent unit and trace every calibration input
Reproducibility failure	Run it in a clean environment from documented instructions
Scientific overreach	Write the next experiment that could falsify the claim

6.6 Evaluation as a curriculum thread

Evaluation is Track 17, but it also appears in every vertical track. Property prediction teaches scaffold and time splits. Structure teaches template and sequence leakage. Omics teaches donor and perturbation units. Agents teach citation verification and tool outcomes. Generation teaches search budgets and oracle exploitation. Active learning teaches policy evaluation. Candidate nomination teaches evidence independence.

This repetition is deliberate. Learners should not reach the capstone with an evaluation vocabulary learned only after they have already internalised invalid examples.

6.7 What this project has measured about itself

The audit of Chapter 10 crosses 13 audited systems and 93 topic units, which materialises 1209 provider-topic cells, of which 591 direct observations and 618 explicit unknowns. The recorded coverage of this project is 93 explained, 93 worked, 66 executable, 50 executable on real data and 37 assessed topics. The audit leaves the claim status of both candidate superlatives at not established, and Chapter 11 reports the independent reading that checked these numbers.

Figure 11 draws the bounds across the thirteen audited systems. AIDD Commons is the first row and its bounds are equal at every threshold because all 93 of its cells are direct observations, while every external row is an interval, because an unobserved cell contributes to an upper bound and nothing to a lower one.

The interval shape of the external rows is the most useful thing in that figure. A provider whose upper bound is 92 and whose lower bound is 4 has not been shown to cover little, it has been shown to have been read at an index rather than at its content. The width of the interval measures the reading, not the provider.



Figure 11: Coverage bounds for thirteen providers across five thresholds, each bound a lower and an upper count out of 93 topics. AIDD Commons is the first row and its bounds are equal at every threshold, 93 explained, 93 worked, 66 executable, 50 executable on real data and 37 assessed, because all 93 of its cells are direct observations. Every external row is an interval, because unobserved cells contribute to an upper bound and nothing to a lower one. The bounds are not confidence intervals and no probability attaches to them.

Figure 12 sets the recorded depths against the independent reading of the same ninety-three topics. The reading agreed with the record on 60 topics, read 33 deeper and read 0 shallower. At the thresholds the reading gave 71 executable, 52 executable on real data and 70 assessed, against the recorded 66, 50 and 37. Chapter 11 reports that reading in full, including what the discrepancy means and why the recorded values were not raised to meet the reading.

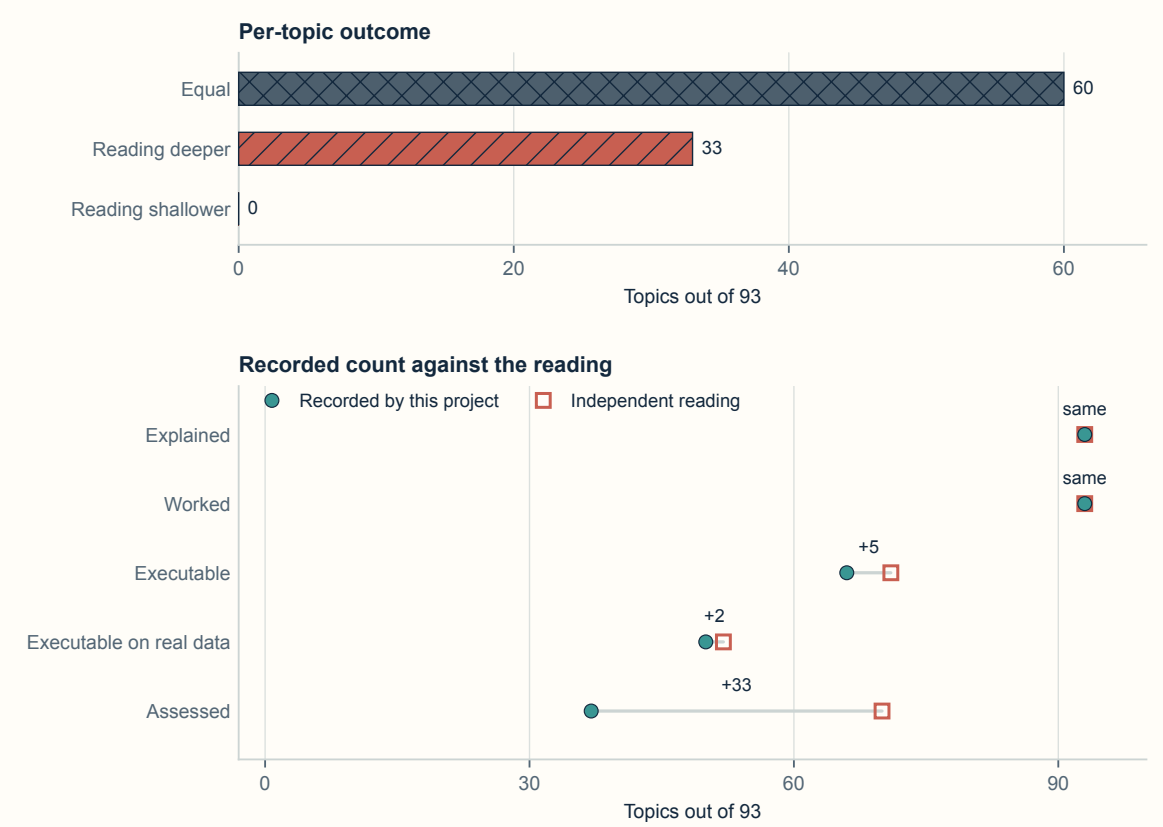


Figure 12: This project's own record against a blind second reading of the same 93 topics by a reviewer from a different model family on 2026-09-11. The reading agreed with the record on 60 topics, read 33 deeper and 0 shallower. At the thresholds the reading gave 93 explained, 93 worked, 71 executable, 52 executable on real data and 70 assessed against the recorded 93 explained, 93 worked, 66 executable, 50 executable on real data and 37 assessed. The reviewer had also written parts of the course, so this is a second reading and not a blind replication.

Figure 13 draws the same ninety-three topics distributed by track and by the basis of each track's runnable work. Five tracks have no runnable topic at all, which the figure shows as an empty bar rather than as a zero, because a track with no runnable topic is a different fact from a track whose runnable topics were not counted.

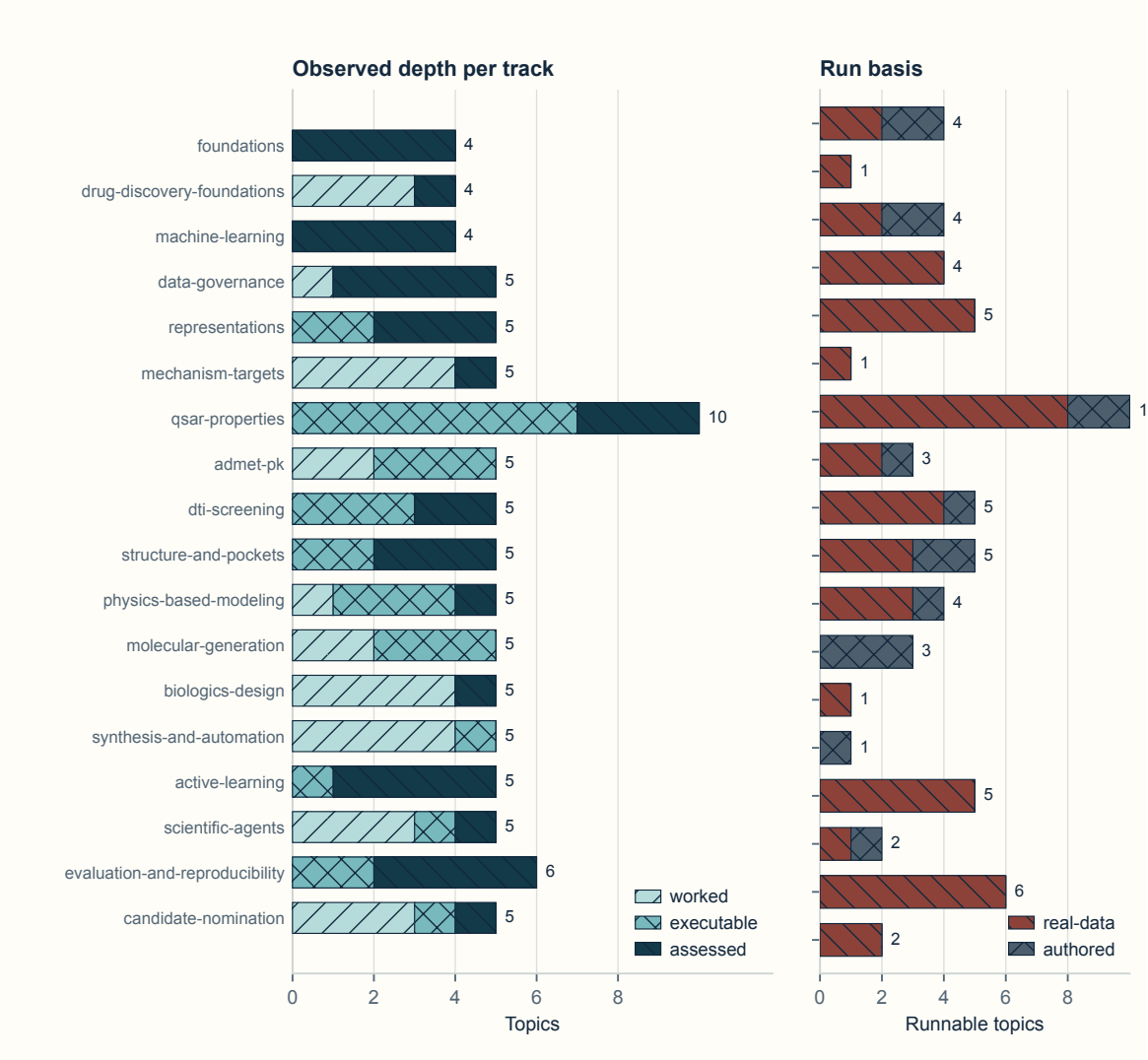


Figure 13: Observed depth per track for this project's 93 audited topics, beside the basis each track's runnable topics rest on. 27 worked, 29 executable and 37 assessed, and of the 66 runnable topics 50 rest on real data and 16 rest on authored inputs. Depth records what a reviewer observed in the material, not what a learner achieved.

Chapter 7

Community development and governance

7.1 Contributions as typed scientific work

AIDD Commons offers six public contribution routes.

A **source proposal** adds or updates a resource with direct evidence and asset-level terms. A **correction** identifies an error and the evidence needed to repair every affected surface. A **lesson proposal** defines a question, prerequisites, learning outcomes, data contract, baseline, and failure modes. A **lab proposal** freezes an executable scientific protocol and expected artifact. An **experiment report** contributes a measured result, replication, negative result, or failure analysis. An **RFC** proposes a material change to scope, schema, curriculum, evaluation, licensing, governance, or safety policy.

Each route asks for evidence, licence and attribution, audience, scientific assumptions, and conflicts. The forms avoid collecting secrets or internal operational records. They are intake structures, not proof that a contribution is correct.

Each route produces a specific artefact when it succeeds. A source proposal produces one reviewed registry record with seven openness dimensions and a licence scope. A correction produces a dated correction that propagates to every affected surface. A lesson proposal produces one lesson reachable from the curriculum graph. A lab proposal produces one executable lab with a retained result. An experiment report produces one reviewed summary with its own stated limits. An RFC produces a recorded decision with its rationale and dissent.

7.2 Editorial review

Review follows the load-bearing claim. A catalog entry is checked against its direct sources and schema. A lesson is checked for conceptual accuracy, prerequisites, examples, and evidence scope. A lab is checked for data identity, split, metric, fit boundaries, expected failure tests, and clean execution. A Chinese page is compared semantically with the canonical English content. A public release is checked for useful contents and exclusion of private workflow material.

Agreement among reviewers is not independent evidence if they read the same source. When a dispute concerns a number, split, licence, or model behavior, review returns to the artifact that defines it. Unsupported claims are narrowed or marked `unverified`, and they are not decided by prestige or majority vote.

The review of a laboratory is the strictest of these, because a lab carries a claim that a number was obtained under a stated procedure. A lab review must establish which entity is one observation and how it was identified, which novelty boundary the partition simulates, which error

in which unit is aggregated over which independent unit, which data were available to the fit and when, which checks the artefact is expected to fail, and whether the run reproduces from the published command.

7.3 Corrections and versions

The project is designed to change. Model availability, repository contents, access terms, and benchmark practice can all move after publication. Records carry fixed last-review dates. Corrections name the affected version and propagate through generated exports, Atlas pages, curriculum prose, and report updates.

Deprecation preserves useful history. A superseded model can remain linked to its successor. A withdrawn dataset can remain visible with its status and warning. Silent deletion would make old lessons and citations harder to understand. Material that cannot lawfully or safely remain is removed with the narrowest public explanation that does not repeat the problem.

A correction that reaches a lab result is the interesting case, because the result has already been published and a reader may have built on it. The correction route requires the evidence needed to repair every affected surface, and the protocol in this project requires that a protocol-invalidating error found after the test set was opened is reported as such rather than folded into a new hypothesis. Chapter 9 gives the labs whose committed summaries carry that kind of disclosure, and Chapter 11 gives the audit corrections that changed the coverage numbers of this project itself.

7.4 Decision rights and conflicts

Contributors propose, reviewers assess bounded evidence, maintainers merge and release, and stewards handle private conduct, security, and biosecurity reports. Material policy changes use RFCs. Routine decisions use documented review and rough consensus, but silence is not consent for a breaking change.

Employment, funding, authorship, investment, or close collaboration can create a conflict. A conflicted expert may provide technical evidence but should not be the sole approver of a disputed ranking, licence interpretation, correction, or enforcement action. Governance cannot make a factual claim true, and it can only decide what the project publishes given the evidence.

The independent reading described in Chapter 11 was designed against exactly this failure mode. The reviewer had also written parts of the course, and the report says so, and the reading is described as a second reading rather than a blind replication for that reason.

7.5 English canonical content and Chinese access

English is canonical because it provides one claim-bearing source for a global technical audience. Chinese pages form a complete onboarding and pathway layer rather than decorative translation. Semantic parity means preserving quantities, negation, uncertainty, causal direction, licence scope, and applicability conditions. It does not require literal sentence order.

As the corpus expands, translation review should use paired sections and terminology notes. Machine translation can assist drafting but cannot approve scientific claims. A correction to canonical content must trigger review of every translated surface that depends on it.

7.6 Accessibility and durable delivery

An open tutorial is not open if it cannot be navigated with a keyboard, read at small viewport widths, distinguished in dark mode, or understood without motion. The site therefore treats visible focus, semantic headings, labelled controls, sufficient contrast, reduced motion, touch target size, and responsive tables as release conditions.

Durability also depends on formats. Canonical prose is plain Markdown, resource records are YAML, exports are JSON and CSV, labs are Python and Jupyter notebooks, and the report is editable Markdown and BibTeX with a PDF derivative. Static site generation avoids an account or application backend. These choices lower the barrier to mirroring, review, and migration.

7.7 Sustainable maintenance

Comprehensiveness is a maintenance problem, not a one-time scrape. The catalog needs scheduled link reports, but link availability is not evidence support. Curriculum coverage needs explicit denominators. Frontier records need dated re-review. Labs need dependency and data-version maintenance without changing frozen historical results. Community health needs correction response and credit for curation, translation, accessibility, and negative results.

The project should grow through evidence-qualified vertical slices rather than thousands of shallow links. A useful slice connects a decision map, core lesson, source records, lab or worked case, evaluation traps, openness analysis, and contribution route. Breadth then grows without sacrificing depth.

The maintenance task differs by artefact, and each one has a unit that shows the work was done. A registry record needs a new dated review and an updated record. A curriculum lesson needs a dated review. A laboratory needs a new environment record with the frozen result unchanged. A comparison record needs a new external-survey date. The audit report needs a new audit date and new bounds. A contribution needs a dated response that names the evidence. This report needs the numbers the audit changed.

Chapter 8

Safety, licences, limitations, and roadmap

8.1 Scientific and biological safety

AIDD methods are dual-use. Property prediction, sequence design, synthesis planning, literature agents, laboratory automation, and optimisation can support beneficial research or lower barriers to harmful work. The educational response should be specific and proportionate. Removing all technical content would undermine legitimate training, and publishing operationally complete harmful workflows would be irresponsible.

Contributions require private escalation when they could materially improve the design, acquisition, optimisation, evasion, or deployment of a biological agent or harmful chemical. Risk increases when a contribution combines actionable targets, optimisation code, operational parameters, and validation guidance. High-level discussion, defensive evaluation, public benchmark analysis, and ordinary molecular modelling are not automatically restricted. Stewards should name the capability change and use the least restrictive adequate response, which may be narrowing scope, delaying details, obtaining expert review, applying access controls, or rejecting the contribution.

Patient and proprietary data create separate risks. Tutorials must not redistribute protected health information, confidential assay data, access tokens, or assets whose terms prohibit sharing. Synthetic examples should not be presented as clinical validation. Public database status does not remove the need to examine consent, governance, bias, and intended use.

8.2 Licence layers

Original AIDD Commons software is licensed under Apache-2.0. Original prose and figures are licensed under CC BY 4.0. Third-party assets retain their own terms. A project-wide licence cannot relicense an external paper figure, dataset, model parameter file, software dependency, trademark, or quoted passage.

Licence review therefore follows the asset. The SPDX licence list is the shared vocabulary the registry uses to record the terms of a third-party asset (Linux Foundation 2026).

- code licence for source and binaries;
- model terms for weights and hosted inference;
- data terms for access, training, transformation, derived artifacts, and redistribution;
- content terms for text, diagrams, screenshots, and educational adaptation;
- citation and notice requirements for scientific and community credit.

An official repository name is not licence evidence. A licence file may cover code but not

weights or data downloaded by the code. "Research use" can restrict commercial or other activity and should not be collapsed into open source. When scope cannot be established from opened terms, the record remains unverified and links readers to the upstream source.

The registry distinguishes six licence layers. Three records fall under a public domain dedication, 101 under a permissive software licence, 19 under a Creative Commons attribution, 22 under a copyleft term, 6 under a Creative Commons non-commercial term, and 6 under other stated terms, and 34 records have no licence recorded at all.

Table 7: Licence layers and what each one covers.

Layer	What it covers	Records
Public domain dedication	The asset may be used without restriction	3
Permissive software licence	Source or binaries, with attribution	101
Creative Commons attribution	Redistribution with attribution, including commercial use	19
Copyleft	Redistribution with the obligation to keep derived works open	22
Creative Commons non-commercial	Redistribution for non-commercial use only	6
Other stated terms	A licence outside the above layers, including platform agreements and research-only grants	6
No licence recorded	The licence was not read, which is not the same as having no licence	34

The layer a record falls in is the most restrictive licence it carries, and twenty-two records carry more than one licence. The research-only layer is the one that most often surprises a reader, because a record can be fully open in every openness dimension and still be unusable for a commercial purpose. The four research-only records in this catalogue are the clearest case of a dimension and a licence carrying different information.

8.3 Limitations of v0.1

The main limitation is coverage maturity. The source registry is broad but not a systematic census. Regional resources, non-English material, emerging companies, proprietary platforms, negative datasets, laboratory automation, clinical translation, and specialised modality communities are likely underrepresented, and resource availability can change after the fixed review date.

The curriculum structure is complete at the track level, but chapter depth and executable lab coverage are uneven. The target-to-candidate and co-folding slices are documentation-rich, while many specialised topics currently provide an orientation and source set rather than a full sequence of lectures, assessments, and labs. Chinese content provides complete onboarding and pathway summaries but not yet a full translation of canonical chapters.

The Atlas's evidence model improves transparency but does not itself verify scientific quality. A directly supported repository claim can still describe a method with weak external validation,

and a closed method may be scientifically strong but impossible to reproduce. Openness and performance are separate axes.

The FreeSolv lab demonstrates protocol discipline on a small molecular regression problem, and its results do not establish general behaviour for medicinal chemistry, large datasets, classification, three-dimensional structure, or distribution shifts. The ChEMBL EGFR lab shows that a representation that wins one frozen comparison can lose another by a wide margin.

The report does not provide an unmatched performance ranking for co-folding or other frontier systems. Such a ranking would require common inputs, databases, templates, inference budgets, evaluation code, exclusions, and independent-unit uncertainty. Public leaderboards that differ on these dimensions cannot be combined responsibly by copying headline numbers.

Each limitation forbids a specific claim. The registry is broad but not a systematic census, so the catalogue is not exhaustive. Many specialised topics carry orientation rather than a full sequence, so the tracks are not equally deep. The Chinese content is a complete pathway layer rather than a full translation, so the Chinese edition does not carry every scientific claim. The evidence model records openness rather than scientific quality, so an open record is not a good method. The FreeSolv result is one frozen split on one small endpoint, so it does not generalise. The ChEMBL EGFR result is two separate comparisons, so no pooled superiority claim is available. No unmatched frontier ranking was produced, so there is no performance ordering of co-folding systems. And seventeen of eighteen tracks contain runnable work while one does not, so complete runnable coverage is not available either.

8.4 Coverage audit required for a superlative claim

Before describing AIDD Commons as the world's most comprehensive tutorial, the project will run a dated comparison audit. The audit will:

1. define a public comparison set of active AIDD curricula, tutorial collections, benchmark hubs, and reference systems;
2. define coverage units for pipeline decisions, scientific topics, resource types, evidence quality, executable labs, languages, accessibility, and maintenance;
3. separate topic presence from substantive treatment and tested execution;
4. archive direct evidence for each comparison entry and record inaccessible pages as unverified;
5. publish the scoring rule before calculating results;
6. invite corrections from compared projects;
7. report both strengths and gaps rather than compressing everything into one score.

That audit has now been run, and its method and results are the subject of Chapter 10. Two candidate claims are on the table, and both read as not established. The claim-promotion gate has a sixth condition, a public correction window that requires publication, and that correction window is not recorded, so the condition is unmet. A bound condition being satisfied is a necessary part of the claim and not a sufficient one.

Until that gate passes, the public wording remains "building the most comprehensive open AIDD learning system". This is a constraint on the project's own claims, and not a reduction in ambition.

8.5 Roadmap

Curriculum depth. Expand every track from orientation into a sequence of foundations, core methods, worked cases, labs, assessments, and capstones. Map each chapter to prerequisites and learning outcomes so that pathways can be checked rather than merely described.

Evidence coverage. Broaden primary-source records across databases, standards, experimental methods, models, industrial platforms, regional education, and historical systems. Add re-review queues for volatile availability and licence claims. Build a public gap ledger rather than inferring absence from failed search.

Labs. Add CPU-first baselines for QSAR, ADMET, drug–target interaction, docking sanity checks, uncertainty, and synthesis data. Add larger optional GPU labs only when a small protocol-valid route cannot teach the claim. Every lab will retain frozen data, split, metric, and selection rules.

Vertical slices. Develop source-to-decision slices for target discovery, perturbational omics, structure and co-folding, virtual screening, multi-objective molecular design, protein and antibody design, retrosynthesis, active learning, and scientific agents. Each slice will include a falsifier and an independent validation route.

Open development. Publish useful experiment reports, including failed replications and engineering limits. Link process decisions to resulting lessons without exporting private operational artifacts. Track changes in inference code, training code, weights, data recipes, training data, and evaluation code independently.

Community and language. Recruit disciplinary editors, educators, software maintainers, translators, accessibility reviewers, and early-career contributors. Extend Chinese semantic parity and evaluate additional language pathways when maintainers can sustain scientific review.

Evaluation and safety. Create known-bad fixtures for leakage, unsupported claims, broken relations, inaccessible interfaces, and release contamination. Develop capability-aware safety review that protects legitimate education while preventing publication of materially harmful operational detail.

8.6 Closing position

The durable contribution of an AIDD tutorial is not a snapshot of fashionable models. It is a public method for learning, checking, and updating the field. That method begins with biological and chemical identity, follows data and decision contracts, demands strong baselines and matched evaluation, distinguishes openness by asset, preserves failures and corrections, and ends with an independently testable claim.

AIDD Commons v0.1 establishes that structure. Its remaining work is visible by design. The project will earn comprehensiveness through maintained evidence, executable depth, and correction, and not through a headline alone.

Chapter 9

The completed laboratory evidence

A result separates from an anecdote by three things. The protocol is frozen before the answer is looked at. The test set is opened once. The artefact is retained, dated, and readable by someone who was not present. Eight laboratory protocols in this repository meet that standard, and this chapter reports what each froze, what the run produced, and what the lab states it does not show.

The point of the chapter is not that the numbers are good. Several are not. The point is that each is a number with a stated procedure behind it, that a reader can check the procedure, and that the negative results are retained next to the positive ones rather than beside them.

9.1 The eight protocols and what they froze

Table 8 lists the eight protocols, the data they read, and the unit their conclusions rest on. The last column is the one that governs how far a result travels, and it is the column that most often turns out to be restrictive.

Table 8: The eight laboratory protocols, their data, and their unit of inference.

Protocol	Source	Unit of inference
FreeSolv v0.52 honest baselines	FreeSolv v0.52, 642 parsed records	Molecule, within a scaffold-disjoint split
ChEMBL 37 EGFR representation	ChEMBL 37, two named assays	ChEMBL parent molecule, within a scaffold group
rMD17 system state	Revised MD17 version 4, two locked systems	Molecular system, not trajectory frame
Experiment pool budget	ChEMBL 37, the same two assays	Acquired compound count against a frozen budget
CYP assay system	ChEMBL 37, two whole targets	Parent molecule measured against both enzymes
AZ clearance system	One AstraZeneca ChEMBL deposition, three systems	Parent molecule measured in two assay systems
LINCS L1000 perturbation context	GEO series GSE70138, two cell lines	Compound, in one cell line against another
Affinity table independent unit	SKEMPI 2.0, a pinned mirror revision	Protein complex, not affinity row

Every one of these protocols was frozen before its test set was opened, and every one states the exclusion rules and the seed that produced them. That shared property is what makes the differences between them comparable.

Three further laboratories carry committed summaries that are descriptive rather than predictive. The decision-unit lab recounts how many independent measurements a compound has, the assay-endpoint lab describes what six archived dose-response assays contain, and the Rosetta lab compares two readouts of the same genetic perturbation. They are reported below under the same discipline, because a recount and a description are still results.

Figure 14 draws the lifecycle that all eight protocols share. The protocol is frozen before the held-out test result is looked at, and the test set is opened once. The sequence runs from the written protocol through the freeze, the structural and finite-output pretests on training and validation data, the single opening, to the reviewed summary. A bug found after the test has been opened is reported as a protocol-invalidating error, and the barred loop is the prohibition on a second opening. The figure carries no measured value, and the two pretests are drawn as they pass rather than as they can fail.

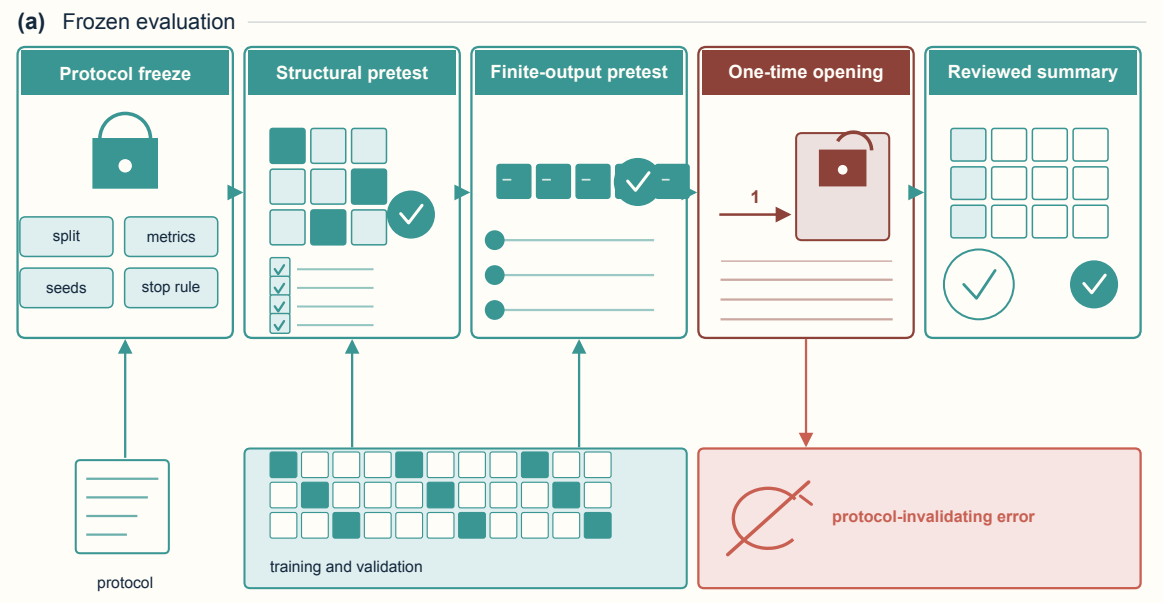


Figure 14: Frozen evaluation. The protocol is frozen before the held-out test result is looked at, and the test set is opened once. The sequence runs from the written protocol through the freeze, the structural and finite-output pretests on training and validation data, the single opening, to the reviewed summary. A bug found after the test has been opened is reported as a protocol-invalidating error, and the barred loop is the prohibition on a second opening. The figure carries no measured value, and the two pretests are drawn as they pass rather than as they can fail.

What separates an artefact from a run is the reviewed summary, the last element in that sequence. Every protocol in this chapter produces one, and each omits the sample membership, the fit traces, and the run state, so that the public record carries the numbers and the decision and not the working files.

9.2 The FreeSolv honest baseline

Chapter 6 reports the FreeSolv result in the context of the split effect it exists to expose. The protocol froze a 642-record dataset into 514 training, 64 validation, and 64 test records under both a random and a scaffold partition, with the scaffold partition as the primary.

The scaffold split gives 1.128 kcal/mol MAE for the ridge against 2.591 for the training mean, and the random split gives 1.019 against 2.767. The random ridge interval reaches a coverage of 0.734 at a mean width of 3.165 kcal/mol, below its nominal target in a sample of 64.

What the lab states it does not show is that the random split is only an easier contrast, that scaffold grouping is an RDKit representation choice, and that the fixed features and untuned models are teaching controls rather than evidence of an optimal predictive model.

Figure 15 draws the same six rows as the figure that separates the point metrics from the interval metrics, with the training-set mean first in each panel because it is the cheap baseline.

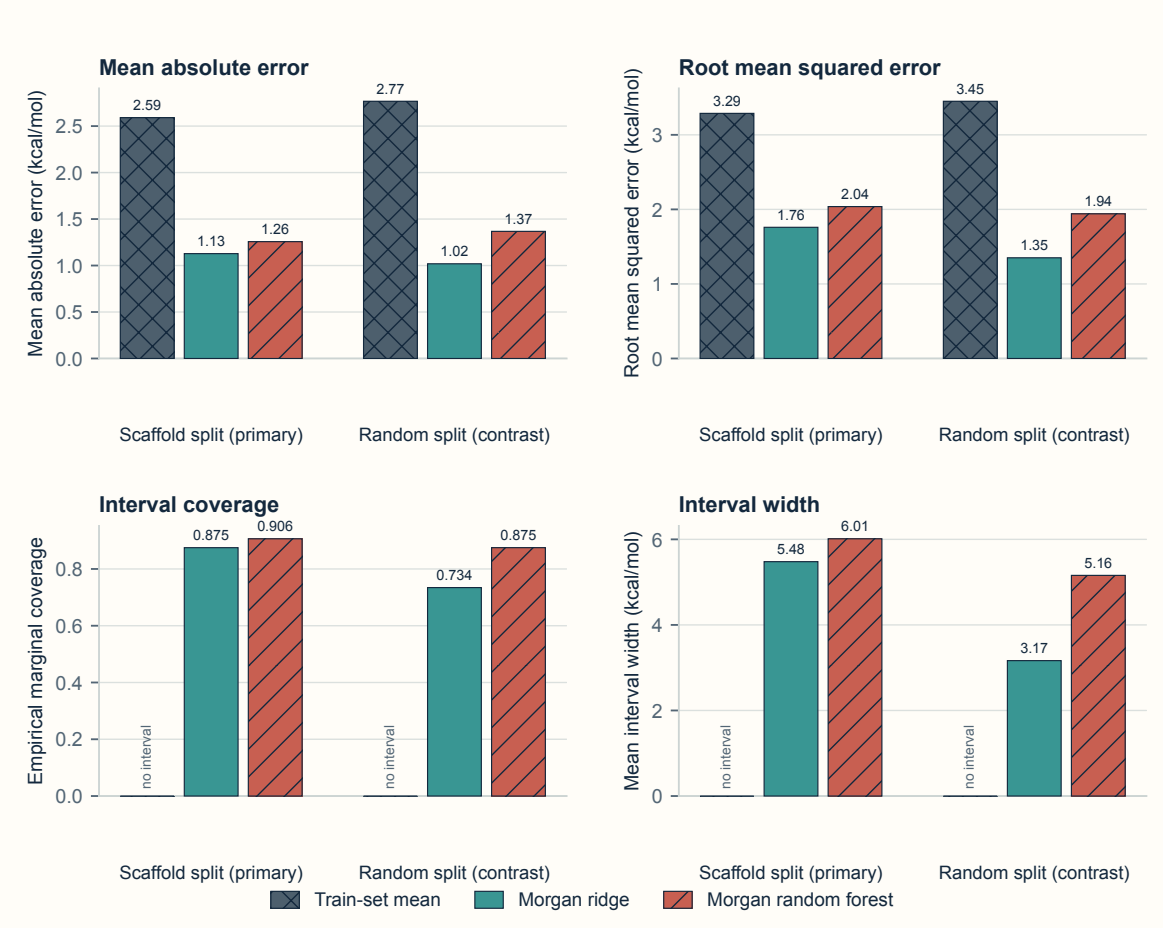


Figure 15: FreeSolv point and interval metrics for two fixed teaching baselines, on the predeclared scaffold split and the easier random split. The ridge reaches 1.13 kcal/mol mean absolute error on the scaffold split against 2.59 for the training-set mean, and 1.02 against 2.77 on the random split. On the scaffold split the intervals reach an empirical coverage of 0.875 for the ridge and 0.90625 for the forest, at mean widths of 5.48 and 6.01 kcal/mol. The coverage is marginal coverage and not conditional calibration, the nominal level of the intervals is not stated in the reviewed results, and these teaching baselines establish no state of the art.

The split comparison is the point of the exercise, and the figure makes it visible. The same model, the same features, and the same fitting code produce 1.128 kcal/mol on the scaffold split and 1.019 kcal/mol on the random split. A number quoted without its split does not identify an experiment.

The lab's own README makes one further point worth repeating. The v0.52 release metadata does not declare licence terms, and the current repository licence files are not version-bound evidence for that release, so the lab does not claim a historical-release licence for the data it read.

9.3 The ChEMBL 37 EGFR representation comparison

The protocol froze the admission rule, the scaffold split, the alpha grid, the selected alphas, and two separate evaluations. The entity is the ChEMBL parent molecule, and the rule requires the molecule identifier to equal the parent identifier, so a salt form is never admitted as a separate observation. The label is the pChEMBL value, aggregated by median across the admitted exact IC50 records in nanomolar units, and the split is by chirality-aware Bemis–Murcko scaffold.

What the run produced is more informative than a single number would be. Table 9 gives the three baselines on the two evaluations.

Table 9: The ChEMBL 37 EGFR representation comparison, on the primary partition and the separate assay replication.

Baseline	Primary MAE (pChEMBL)	Primary Spearman	Replication MAE (pChEMBL)	Replication Spearman
Training mean	0.479	Undefined, constant prediction	0.898	Undefined, constant prediction
Descriptor ridge	0.469	0.129	0.579	0.113
Morgan ridge	0.418	0.505	3.132	-0.041

Figure 16 draws the two evaluations as a slope graph, so that the reversal of the ordering is the visible object rather than something a reader has to compute.

The Morgan ridge is the best of the three on the primary partition of 132 compounds at 0.418 pChEMBL, and the worst of the three on the separate assay replication of 170 compounds at 3.132, where its rank correlation falls from 0.505 to -0.041. A representation that wins one of these comparisons is not uniformly best, and the ordering is a property of the two comparisons rather than of the representation.

What the lab states it does not show is the part that constrains the use of the result. The acquisition lab had accessed labels shared with this evaluation before it ran, so this is not an entirely untouched independent confirmatory evaluation, and the frozen selections were not retuned after that exposure. The two evaluations are separate comparisons, and success on one does not erase failure on the other. The descriptor alpha reached the upper edge of the frozen grid, which means the search did not bracket the optimum and the reported figure understates what that baseline can do. No confidence intervals, p values, or pooled-assay superiority claims are reported.

9.4 The rMD17 system state

The protocol read the revised MD17 release, figure 4, under a licence that permits redistribution, and selected four systems by declared role, with benzene and uracil as the locked systems and ethanol and malonaldehyde as the development systems. It selected 1000 frames per system by a stated rank expression and partitioned them into 600 training, 200 validation, and 200 evaluation frames. The features are element-pair distance bins from 0.0 to 10.0 Angstrom in 0.1 steps, and the protocol forbids the forces and old-energy arrays as features, so a model cannot read the answer from the trajectory.

What the run produced is in Table 10. Each model is fitted on its own system's training frames.

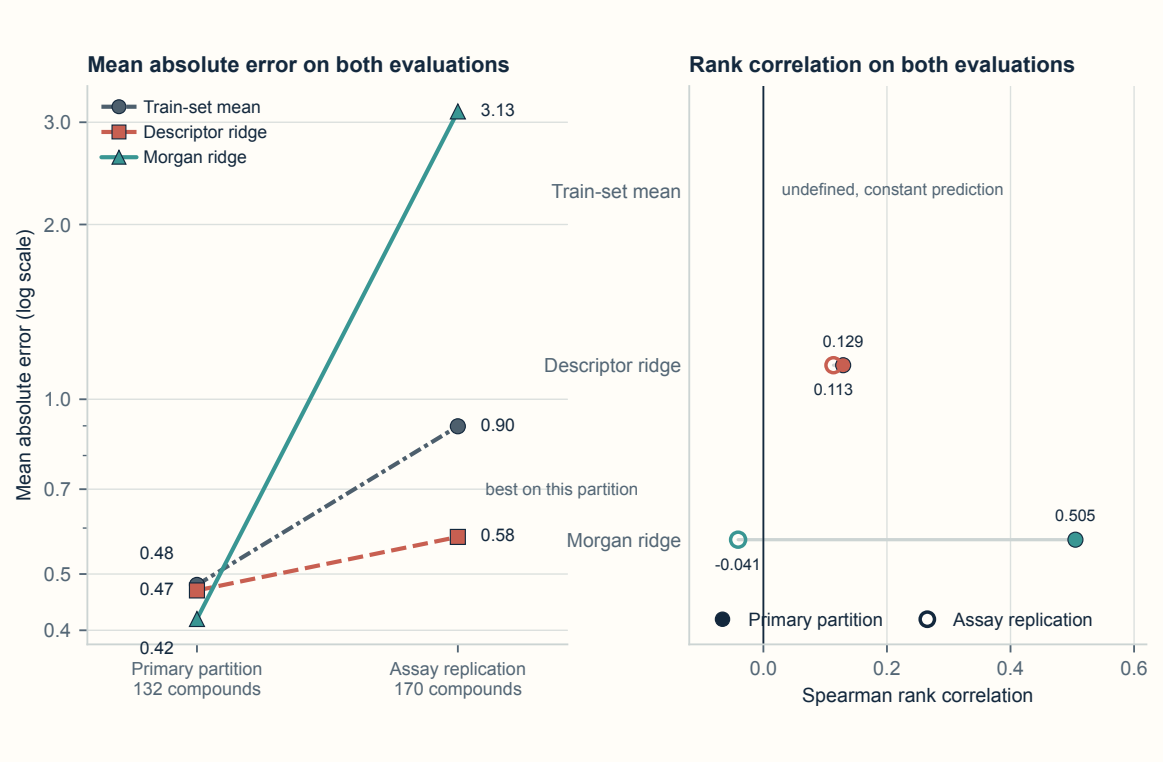


Figure 16: The same three baselines on the primary partition of 132 compounds and on the separate assay replication of 170 compounds. The Morgan ridge is the best of the three on the primary partition at 0.418 mean absolute error, while the descriptor ridge is the best on the replication at 0.579; the Morgan ridge reaches 3.132 there and its rank correlation falls from 0.505 to -0.041. The two evaluations are separate comparisons and success on one does not erase failure on the other, the acquisition lab had accessed labels shared with this evaluation before it ran, and no confidence intervals or p values are reported.

Table 10: The rMD17 system-state result for the two locked systems.

System	Control MAE (kcal/mol)	Ridge MAE (kcal/mol)	Control RMSE	Ridge RMSE	Target range
Benzene	2.234	1.602	2.911	2.055	16.805
Uracil	3.720	3.124	4.784	4.087	33.140
Median across systems	2.977	2.363			
Range of per-system MAE	1.486	1.522			

What the lab states it does not show is that each model is fitted on its own system's training frames, so this is not zero-shot transfer. The reporting units are two molecular systems and frames within each system are correlated, so these are within-system evaluations and not 400 independent observations. No frame-pooled metrics, confidence intervals, p values, force, or state-of-the-art claims are reported. The original evaluation did not retain per-frame Ridge predictions, so the post-run verification recomputed the training-mean scores and target ranges rather than the Ridge scores.

The correlated-frame problem is severe enough that a separate lab page was written for it. Figure 18 draws the naive standard error of a trajectory observable next to the block-averaged one.

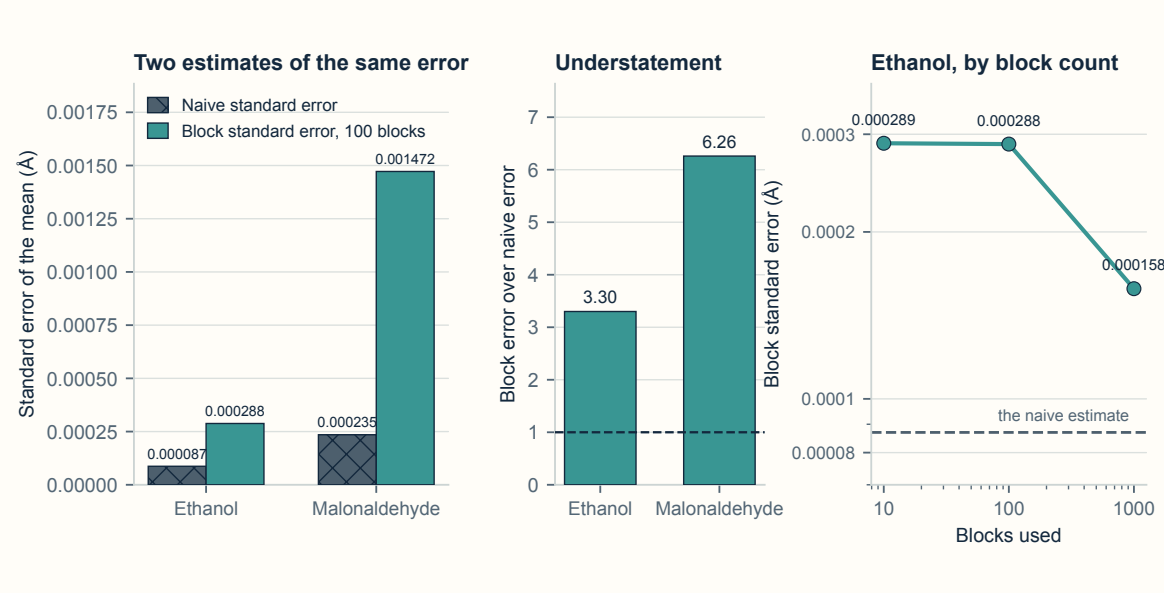


Figure 17: The uncertainty of a trajectory observable, from the rMD17 lab page. Ethanol has a naive standard error of 0.000087 Å against a block-averaged standard error of 0.000288 Å at 100 blocks and Malonaldehyde has a naive standard error of 0.000235 Å against a block-averaged standard error of 0.001472 Å at 100 blocks. The middle panel is the ratio between the two estimates, which is the number the page reports. The right is ethanol's block-averaged standard error against the block count, because a corrected estimate still carries a choice inside it. There is one trajectory per molecule and one level of theory, so nothing here speaks to replicate dependence or force-field sensitivity.

The naive standard error understates the uncertainty of the estimate by a factor of 3.30 for ethanol and 6.26 for malonaldehyde, and the right-hand panel shows that the corrected estimate still carries a choice inside it, since the block count changes the answer. One trajectory per molecule and one level of theory means nothing in the figure speaks to replicate dependence or force-field sensitivity.

Figure 19 draws the convergence of the same observable from the other side, as the cumulative mean of the radius of gyration against the full 100,000-frame mean.

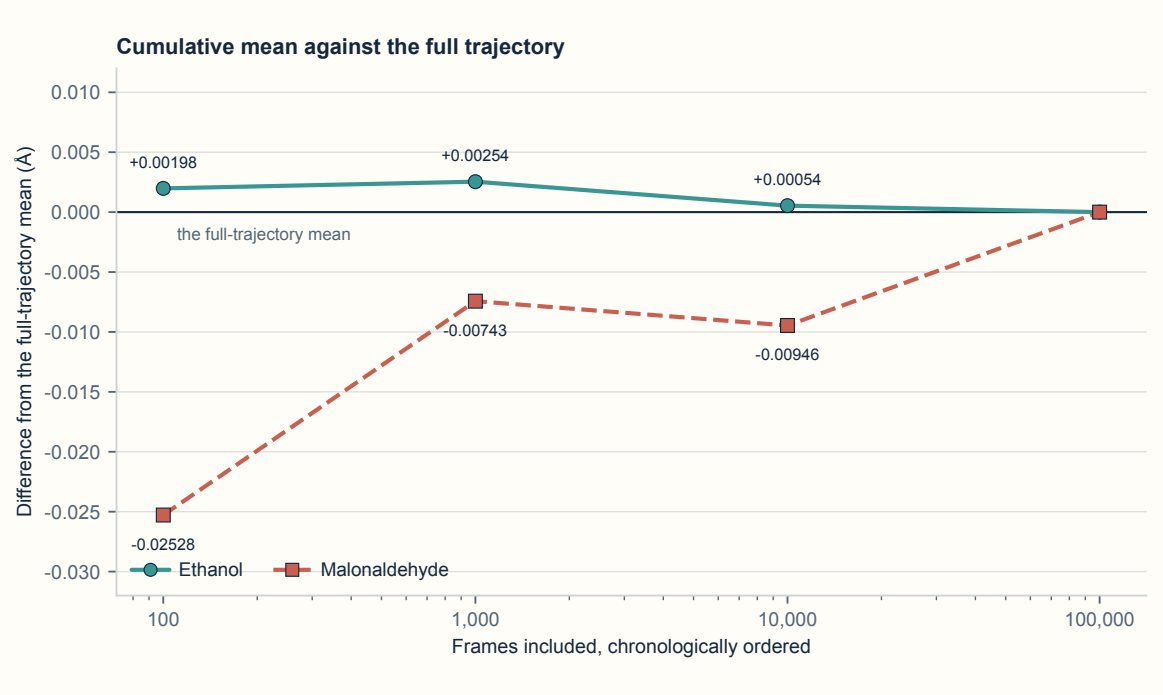


Figure 18: The cumulative mean of the radius of gyration against the full 100,000-frame mean, from the rMD17 lab page. +0.00198 Å for ethanol and -0.02528 Å for malonaldehyde at 100 frames, +0.00254 Å for ethanol and -0.00743 Å for malonaldehyde at 1,000 frames and +0.00054 Å for ethanol and -0.00946 Å for malonaldehyde at 10,000 frames. The malonaldehyde curve is not monotone, being closer at a thousand frames than at ten thousand, and that is drawn as the data stands. One trajectory per molecule and one level of theory means nothing here speaks to replicate dependence or force-field sensitivity.

The malonaldehyde curve is not monotone, being closer at a thousand frames than at ten thousand, and the figure draws that as the data stands. A reader who checked only the ten-thousand-frame point would conclude that more frames made the estimate worse, and the curve is the only thing that shows the conclusion is wrong. Ten percent of a trajectory is not automatically enough, and a fixed number of frames does not transfer to a different system.

9.5 The experiment pool budget

The protocol is a retrospective teaching specification, and it says so. It reuses the frozen ChEMBL EGFR endpoint, parent-record, median-aggregation, and filter contracts without relaxing any filter. It partitions the parent records into an initial set, a pool, and an evaluation set by whole scaffold groups, with target capacities of 10, 70, and 20 percent. The budget is four rounds of the floor of the pool size divided by twenty, and the realised fraction is reported rather than described as exactly 20 percent. The target universe is the original pool only, and recovery is the fraction of the top decile acquired at each checkpoint, integrated over the checkpoints and divided by the final acquired count.

The evaluation model is fixed at Ridge with alpha 1.0, and the protocol declares the tuning forbidden. At checkpoint zero the model is fitted on the initial parents only and the complete evaluation prediction vector is locked before any evaluation label is scored.

What the run produced is in Table 11. The two named policies and the five fixed random instances are all visible.

Table 11: Acquisition policy recovery on two retrospective assay replays.

Assay	Pool	Acquired	MaxMin recovery AUC	Ensemble-variance recovery AUC	Range over five random instances
CHEMBL5736995	631	124	0.0508	0.1133	0.0762 to 0.1016
CHEMBL3705421	130	24	0.0000	0.1635	0.0096 to 0.1058

Figure 17 draws the recovery curves and the evaluation error at the same checkpoints, with the two named policies in the categorical colours and the five random instances in one colour with a distinct marker each, because they are five fixed seeds of one policy and not five different methods.

The diversity policy recovers less of the top decile than every one of the five random instances on the primary assay, and it recovers none of it on the second assay. That is a negative result and it is retained as such. The second assay has a pool of 130 parents in one scaffold group, which makes its diversity fraction uninformative, and the ensemble-variance policy wins it for a reason that has nothing to do with ensemble variance.

What the lab states it does not show is that these are two retrospective assay replays and not prospective experiments or discovery claims. The random seeds are fixed policy instances and not independent biological replicates. The Ridge alpha of 1.0 was informed by the earlier development work on the same contract, so this run is not an independent corroboration of that work, and the acquisition lab had accessed labels shared with the earlier evaluation before that evaluation ran, which this run inherits. No confidence intervals, p values, or pooled inferential comparisons are reported, and the protocol forbids a reader from concluding prospective synthesis, assay completion, hit confirmation, a new biological discovery, or population inference from random seeds.

9.6 The CYP assay system

The protocol reads two whole ChEMBL 37 targets rather than two named assays, because a single CYP target's IC50 record set is itself spread across documents. It reuses the admission rule of the ChEMBL EGFR protocol unmodified, and defines a clean single pair as a parent with exactly one admitted activity per target.

What the run produced is in Table 12. The correlations are computed once over the full overlap and once over the clean single-pair subset, as a robustness check of each other, and no model of any kind is fitted anywhere in the protocol.

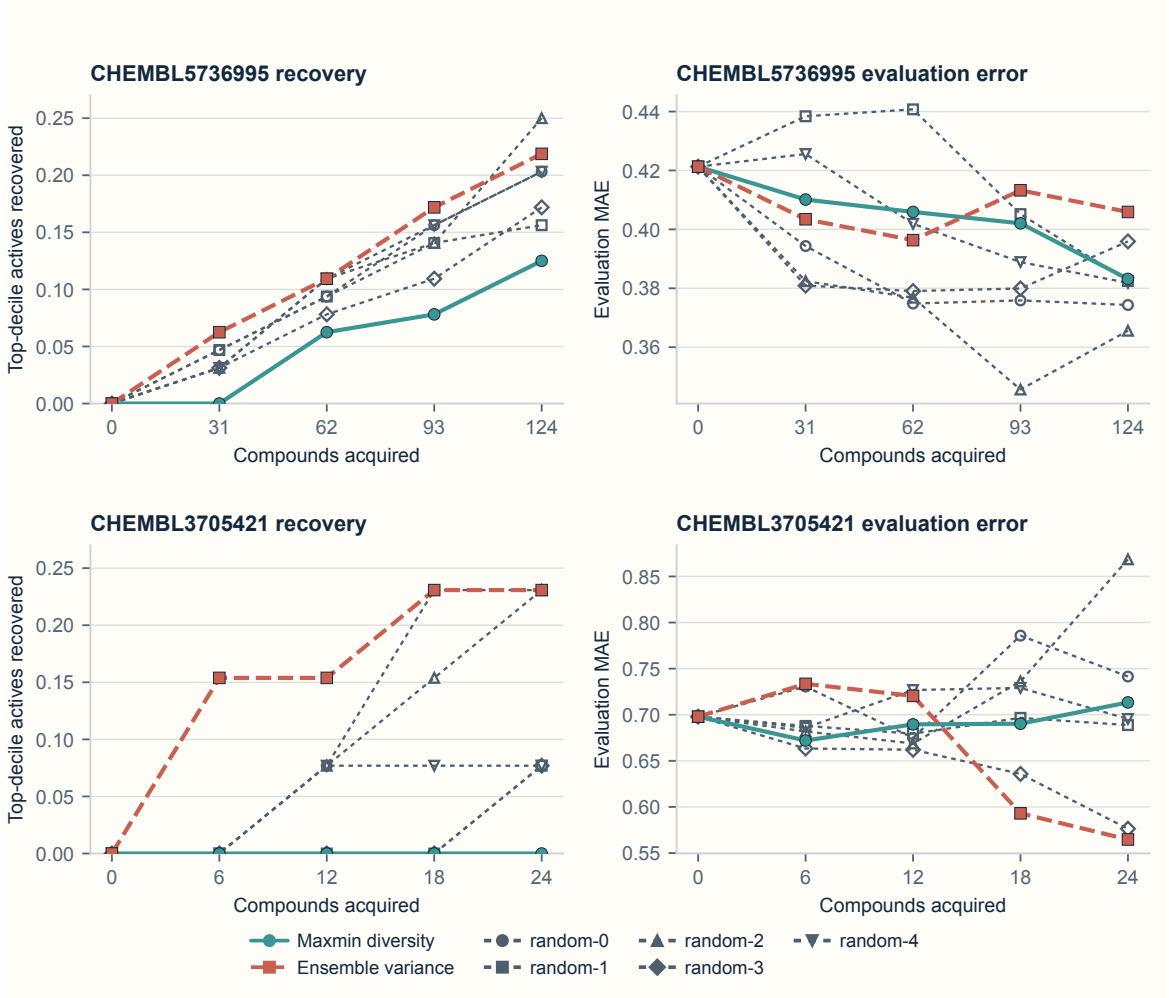


Figure 19: Acquisition policies on two retrospective assay replays, on CHEMBL5736995, 124 compounds were acquired against a pool of 631, and the diversity policy recovered 0.0508 of the top decile against 0.1133 for the ensemble-variance policy and on CHEMBL3705421, 24 compounds were acquired against a pool of 130, and the diversity policy recovered 0.0000 of the top decile against 0.1635 for the ensemble-variance policy. The two named policies are drawn in colour and the five random policy instances in one colour with a distinct marker each, because they are five fixed seeds of one policy rather than five methods. These are retrospective replays and not prospective experiments, the random seeds are not independent replicates, and no confidence intervals or p values are reported.

Table 12: The CYP assay-system comparison and the AZ clearance-system comparison.

Pair	What varies	Shared molecules	Pearson r	Spearman rho
CYP2D6 against CYP3A4, all pairs	The enzyme	1195	0.299	0.298
CYP2D6 against CYP3A4, clean pairs	The enzyme	1015	0.299	0.304
Microsome against rat hepatocyte	Species and matrix	359	0.341	0.387
Microsome against human hepatocyte	Matrix, species fixed at human	96	0.408	0.486
Human hepatocyte against rat hepatocyte	Species, matrix fixed at hepatocyte	106	0.310	0.473

The two systems admit 5251 CYP3A4 parents and 2781 CYP2D6 parents, of which 1195 are measured against both.

What the lab states it does not show is that the protocol reads only in vitro IC50 inhibition records. It says nothing about transporters, enzyme induction, metabolite identification, or a clinical drug-drug-interaction outcome for any named drug pair.

Figure 20 draws the pairs and the largest disagreements between them.

A correlation near 0.3 between two assay systems means that a measurement in one system carries little information about the measurement in the other, and that is not the same as saying either system is wrong. The six molecules drawn are the extremes the review found rather than a sample of the distribution, and they span 2.46 to 2.90 pChEMBL in both directions, with the first system higher on three and the second on three.

9.7 The AZ clearance system

The protocol reads one ChEMBL dataset deposition that three systems share, so there is no separate document condition in the overlap contract. It reuses the parent-structure and duplicate-median admission discipline of the two sibling protocols, adapted to a real difference in this endpoint. Intrinsic clearance reports an exact value with an empty relation field and uses the inequality only for a reading outside the assay's quantifiable range, so the admission rule requires the relation to be absent rather than equal.

The three systems are human liver microsomes, rat hepatocytes, and human hepatocytes. They admit 743, 592, and 288 parents respectively, and no parent is a duplicate within a system.

What the run produced is the lower half of Table 12. Every pairwise correlation is between 0.31 and 0.49.

What the lab states it does not show is that the protocol reads only in vitro intrinsic-clearance records, one of several measures its topic names. It says nothing about absorption, volume of dis-

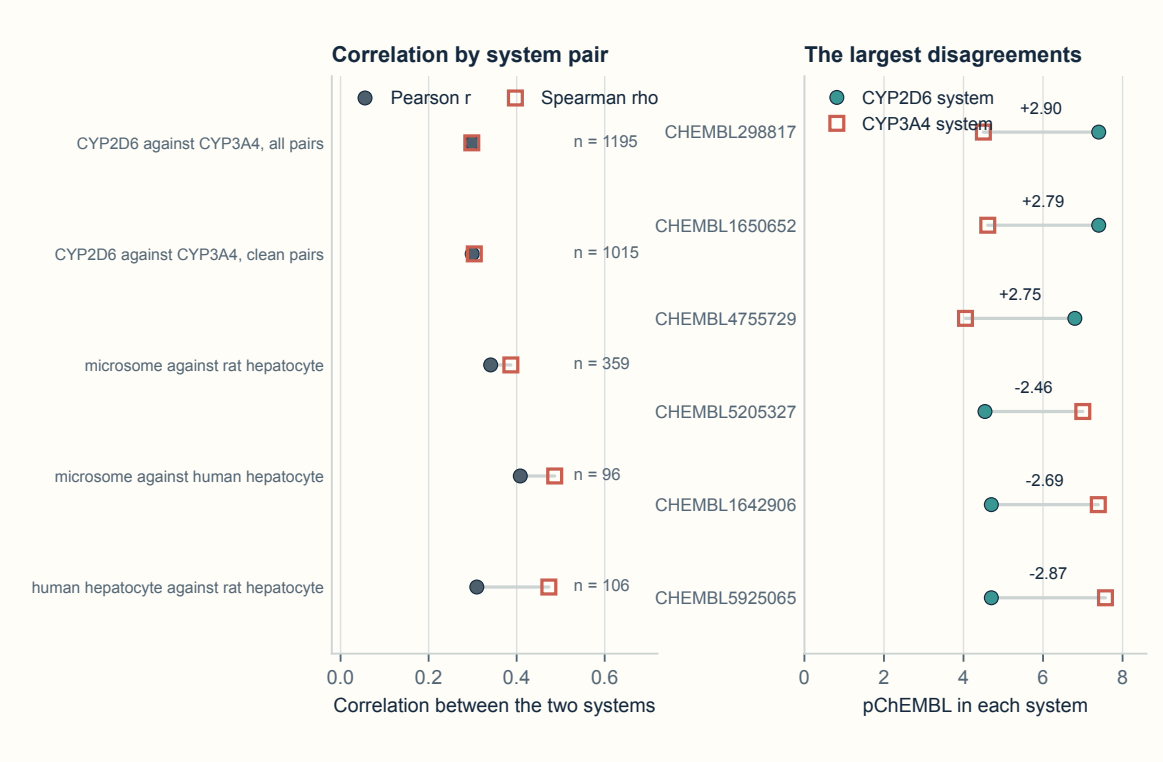


Figure 20: Two pairs of assay systems that disagree. CYP2D6 against CYP3A4, all pairs reaches a Pearson r of 0.2987 and a Spearman rho of 0.2981 over 1195 shared molecules, CYP2D6 against CYP3A4, clean pairs reaches a Pearson r of 0.2990 and a Spearman rho of 0.3038 over 1015 shared molecules, microsome against rat hepatocyte reaches a Pearson r of 0.3410 and a Spearman rho of 0.3868 over 359 shared molecules, microsome against human hepatocyte reaches a Pearson r of 0.4083 and a Spearman rho of 0.4864 over 96 shared molecules and human hepatocyte against rat hepatocyte reaches a Pearson r of 0.3095 and a Spearman rho of 0.4732 over 106 shared molecules. The six largest disagreements between the two CYP systems span 2.46 to 2.90 pChEMBL in both directions, with the first system higher on three molecules and the second on three. A correlation near 0.3 means a measurement in one system carries little information about the other, which is not the same as saying either system is wrong, and the six molecules are the extremes the review found rather than a sample of the distribution.

tribution, or oral bioavailability, and makes no causal or human-dose claim. An in vitro clearance disagreement between two assay systems is not a prospective interaction or dosing statement.

One methodological catch belongs here because it constrains what the numbers can be compared against. Microsome values are reported per gram of protein and hepatocyte values per million cells, so only the rank correlations compare validly across a matrix change, and a raw fold-change between them does not.

9.8 The LINCS L1000 perturbation context

The protocol reads GEO series GSE70138 and fixes exactly two cell lines, one dose, and one timepoint out of the several thousand the same series contains, and it uses only the 978 directly measured landmark genes. Each admitted profile has the median profile of every vehicle-control well on that exact plate subtracted before aggregation, and an instance whose plate has no vehicle-control well is excluded rather than paired with a control from a different plate.

What the run produced is a two-stage result that is the most instructive in this chapter. The cell lines admit 1725 compounds from 37495 input rows for MCF7, and 1724 from 34112 for PC3, of which 1721 are shared. Computed once on raw Level 2 values with no baseline correction, the matched compound pairs show a median Spearman rho of 0.79 against a mismatched-pair null median of 0.70. After vehicle-control subtraction the matched median is 0.175 against a null median of 0.034, a difference a two-sided Mann-Whitney U test finds significant at $p = 7.9 \times 10^{-82}$.

Table 13 gives the two stages of the comparison, and the first row is the one a reader should carry.

Table 13: The L1000 cross-cell-line comparison before and after baseline correction.

Comparison	Matched median rho	Mismatched null median rho	Matched fraction positive	Null fraction positive
Raw Level 2 values, no baseline correction	0.79	0.70	not reported	not reported
After per-plate vehicle-control subtraction	0.175	0.034	0.795	0.603

What the lab states it does not show is that the protocol reads only two real cell lines, one dose, and one timepoint out of the several thousand the same GEO series contains, and only the 978 directly measured landmark genes. It makes no claim that a cross-cell-line signature correlation predicts any clinical outcome.

The first row of Table 13 is the finding a reader should take away, and it was caught before publication rather than after. On raw values, a compound's profile in one cell line correlates with an unrelated compound's profile in the other cell line almost as well as with its own, because both cell lines' raw profiles share enough baseline structure. A method that skipped the baseline correction would have reported a strong agreement signal and been measuring the cell lines rather than the compound.

The comparison still has a residue that the numbers state plainly. About one-fifth of the real compounds show a negative cross-cell-line correlation even after baseline correction, so cell-line

context measurably changes and, for a real minority of compounds, reverses the direction of a compound's transcriptional response.

9.9 The affinity table independent unit

The protocol reads a pinned mirror of the SKEMPI 2.0 table (Jankauskaitė et al. 2019), and its declared question is how many independent measurements a compound actually has and which rows have to be set aside before that number means anything. The entity is one upstream record, the independent unit is the complex, and the analysis subset is the antibody-antigen class with both parsed affinities finite and positive. The split sorts distinct complexes, shuffles with the fixed seed, and reserves the first ceiling of 20 percent of complexes for test.

What the run produced is in Table 14.

Table 14: The SKEMPI complex-holdout result.

Comparator	Test-row MAE (log10 M)	Test-complex mean MAE (log10 M)	Character
Training-group median only	1.427	1.684	Uses no observed mutant value
Observed mutant as the wild-type prediction	1.132	1.560	An observed comparator, not a prospective predictor

The holdout is 9 test complexes holding 33 test rows, drawn from 35 training complexes and 686 training rows. The table holds 2486 complete pairs across 348 complexes, of which 4669 rows carry no class and 5739 rows are fragmented records. A diagnostic row split leaks 142 of 144 test rows into training, which is the measurement the protocol was designed to make visible.

What the lab states it does not show is that the primary-value rule deliberately does not reconstruct overflow fragments, so a partial mirror record is not evidence that an upstream measurement was never made. The split demonstrates the independent-unit issue for this table, and two simple comparators do not establish model superiority. No reader outcomes have been observed.

9.10 The three descriptive laboratories

Three laboratories in this repository produce a description rather than a prediction, and they are reported here because a reader who saw only the predictive laboratories would form a wrong account of what a laboratory is in this project.

The decision-unit lab asks, for a declared set of real compounds, how many independent measurements each compound actually has, and which rows have to be set aside before that number means anything. The protocol draws no split and holds nothing out, because it fits no model. The cohort is read end to end and every count is published. The grouping unit is the triple of assay, document, and endpoint type, so two admitted point measurements sharing all three are the same measurement.

What the run produced is an admission funnel over 13305 reported rows across ten parent molecules. Of those, 1004 are admitted in total, 516 as bounds and 488 as point measurements, and the point measurements carry 484 independent experiments, of which 71 are salt-form and 417 parent-form measurements. The largest gap between a reported row count and an independent measurement count is valproic acid, with 6294 reported rows and 27 independent measurements,

a ratio of 233 rows per measurement. The exclusion is where the number comes from, and the largest classes are 6363 rows excluded as non-standard activity and 5517 excluded as outside the endpoint family.

What the lab states it does not show is that the cohort is fifteen molecules chosen to exercise the counting rules, and not a random sample of ChEMBL. No uncertainty interval is reported, and none should be, because every number is an exact count of a fully enumerated declared set rather than an estimate of a population quantity.

The assay-endpoint lab asks what six archived PubChem BioAssay dose-response assays actually contain, and which conclusions each assay's own control record licenses. The protocol makes one deterministic descriptive pass over the cached archive, audits the control record of each assay, and recomputes or refits the concentration-response curve. It fits no predictive model, and its evaluation is none, because there is no prediction and no metric that compares a model against a baseline.

What the run produced is a mix of 1529 readouts across five endpoint types and 1257 relations, split into 644 development and 613 held-out compounds by a seeded rule that groups on the PubChem compound identifier. The relation mix is 832 equalities, 406 inequalities, 4 strict inequalities, and 15 unstated. The curve check recomputes each archived curve under a fixed tolerance of 5 points and compares the recomputed endpoint against the value the archive itself recorded, and the fraction of held-out curves within tolerance runs from 0.62 for assay 519 to 0.94 for assay 1277.

What the lab states it does not show is that six assays are not a sample of the archive, and nothing in them estimates how relation qualifiers, confidence scores, or control regimes are distributed across PubChem BioAssay as a whole. A reader who changes the admission rule gets a different cohort. A missing control record is reported as a fact about the archive rather than as evidence that the experiment was performed without controls, and the curve check is a check on the archive's internal consistency rather than an assessment of the underlying experiment.

The Rosetta lab asks how strongly two readouts of the same genetic perturbation agree, and how much of that agreement survives when the plate and batch structure of the measurement is accounted for. The source is one batch of one study, one cell line, one timepoint, with 291 imaging features and 978 transcriptome features over 528 admitted perturbations.

What the run produced is a large apparent agreement and a much smaller robust one. With the split drawn at random across the batch, the cross-modal feature correlation is 0.826. When the batch is blocked out of the training side it falls to 0.633, and when a single plate is blocked out it falls to 0.652 for imaging and 0.669 for transcriptome. A size-matched random fold of the same training and test sizes recovers 0.788 to 0.809, which is the measurement that identifies what the blocking removed. The survival ratio, the share of the above-null agreement that remains, is 0.765 for the batch-blocked evaluation and 0.788 and 0.810 for the two plate-blocked evaluations.

The model-free check reaches the same conclusion without a model. The neighbourhood concordance of the two feature spaces is 0.745 against a permuted null of 0.0007, so the two readouts genuinely describe the same perturbations, and the variance attribution shows the imaging batch accounts for 0.4 percent of the imaging feature variance and the imaging plate for 2.7 percent. The agreement is real and is mostly not caused by the plate structure.

What the lab states it does not show is that this is one batch of one study, one cell line, one timepoint, one set of engineered edits, and that the study's other four batches, the other 46 gallery studies, and the JUMP pilot are outside the protocol. The agreement is measured between two readouts of the same perturbation, and it is not evidence about mechanism, about the gene, about a target hypothesis, or about any clinical outcome. The blocked evaluations hold out structure inside one published experiment, and a batch or plate held out here is still one the laboratory has seen before, so this is not a deployment setting.

9.11 What the laboratory evidence establishes

Table 15 gathers the eight protocols with the single most important limitation each one states.

Table 15: The stated limit of each laboratory.

Laboratory	The limit its own summary states
FreeSolv	The random split is an easier contrast, and the fixed models are teaching controls rather than an optimal predictive model
ChEMBL 37 EGFR	The acquisition lab had accessed shared labels before the evaluation ran, and the two comparisons are separate
rMD17 system state	Two within-system evaluations, not zero-shot transfer, and frames within a system are correlated
Experiment pool budget	Retrospective replay, fixed seeds rather than replicates, and an inherited alpha
CYP assay system	In vitro inhibition only, with no clinical interaction claim
AZ clearance system	One of several clearance measures, with no absorption, distribution, or bioavailability claim
LINCS L1000	Two cell lines, one dose, one timepoint, and no clinical or in vivo claim
Affinity table	A demonstration of the independent-unit issue, and two comparators rather than a model ranking

Three conclusions follow, and none of them is a performance claim. First, the strongest result in the set is the ChEMBL EGFR reversal, because a ranking that flips between two frozen comparisons is evidence about the comparison rather than about the representation. Second, the most useful number in the set is the L1000 raw-correlation figure of 0.79 against a null of 0.70, because it shows how a method can measure the wrong thing convincingly. Third, every one of the eight is bounded by a stated unit of inference, and a reader who takes any of these numbers outside its unit is not extending the result but replacing it.

Each laboratory also states its limits in the artefact itself rather than in a separate note, which is what allows an independent reviewer to check the claim without asking the author what the author meant. That property is what the audit of Chapter 10 and the reading of Chapter 11 both depend on.

Chapter 10

How the comparison audit works

10.1 What the audit is for

The audit exists to make the project's own coverage claim falsifiable by someone who does not trust the project. It does that by fixing a comparison set, a unit of coverage, a threshold, an evidence rule, and a date before any number is computed, and then by publishing the resulting bounds whether or not they favour the project.

That design has a property that a coverage score does not. A score compresses a set of measurements into one number, and a reader cannot tell whether the number came from a wide measurement or a narrow one. The audit keeps the measurement wide. Every provider is crossed with every topic, and the result for each provider is a pair of numbers, a lower bound and an upper bound, at each of five thresholds.

10.2 The frozen comparison set

The audit contains thirteen providers, eighteen tracks, and ninety-three topics. The thirteen providers fall into four classes.

The class matters because the four kinds of system are not measured by the same yardstick, and the audit does not pretend they are. A curriculum that covers a topic as a worked case and a tutorial collection that covers it as a notebook script are different deliverables, and a benchmark hub that lists a task is different again from a system that teaches it. The class is recorded on every row of the matrix, and a reader who wants to compare a curriculum against another curriculum and a tutorial collection against another tutorial collection can do that from the same table.

The 1209 provider-topic cells are the product of 13 providers times 93 topic units. Of those, 591 direct observations and 618 explicit unknowns.

10.3 The lower and upper bound method

Every provider is crossed with every taxonomy topic, and every cell takes one of three states. A direct observation takes precedence over the record's explicit default of unverified. An unverified cell contributes zero to the lower bound and one to the upper bound. An absent cell contributes zero to both. A directly observed cell whose depth reaches the threshold counts in both bounds, and a cell that was observed but sits below the threshold counts in neither, and it lowers that provider's ceiling.

The asymmetry is the whole method, and it is worth stating in plain terms. Not knowing whether a provider covers a topic is not the same as knowing that it does not, so an unknown cell is allowed to raise a ceiling but never to lower a floor. A cell that was opened and found to be

below the threshold is a measurement, and it lowers the ceiling, because a measurement below the threshold is a fact about the provider at the date of the survey.

That second rule is the one that a reader should check, because it is the rule that determines whether the comparison favours the project or the comparator. The audit records that comparator ceilings at the explained threshold were lowered by 364 cells that the project reviewed and found below that threshold, across twelve providers. Had none of them been reviewed, the strongest external upper bound would have been 93 rather than 92, and the project's own lower bound of 93 would not have exceeded it. The margin in the most comprehensive claim is a margin produced by this project's own survey work, and the audit says so in the same paragraph that reports the margin.

10.4 The run basis distinction

Every executable or assessed cell also records what its runnable task rests on. A real-data basis means measured or computed records that exist independently of the course, or a genuine simulation, docking, or structure-prediction run on them. An authored basis means hand-written or synthetic inputs, including real molecules with invented values. An unverified basis means the record does not establish which.

The executable-on-real-data threshold counts only real-data cells. An unverified basis adds to the upper bound alone. This is what keeps a hand-written exercise from counting as a measurement, and it is why the reported coverage carries two executable numbers rather than one.

External cells were classified only from what each cell's own note, reviewed scope, and evidence titles state. No provider was re-read in order to classify it. That is a deliberate limitation with two consequences. The external classification is a reading of the recorded evidence rather than a fresh survey, so it is dated differently from this project's own row. And the external rows therefore carry an interval, because a classification made from a note rather than from the material itself is a weaker observation than a direct one.

10.5 The scoring rule published before the results

The audit publishes the scoring rule before it computes the results. The rule is the bound method above, the threshold set of explained, worked, executable, executable-on-real-data, and assessed, the run-basis definition, and the treatment of unverified and absent cells. It is stated in the audit document and implemented in one place in the exporter, and the exporter refuses to run if a record violates it.

The status field of the two candidate claims is a hardcoded literal in the exporter rather than a computed value. It cannot become established because the numbers moved. A human has to change the code. That is a safeguard and it is deliberate.

10.6 What the audit found

Table 16 gives the coverage bounds for all thirteen providers at the five thresholds. Every external row is an interval, and the intervals are wide because the external survey was a survey of surfaces at a stated depth.

Table 16: Coverage bounds for the thirteen audited systems, out of 93 topics.

Provider	Direct cells	Absent	Unknown	Explained	Worked	Executable	On real data	Assessed
AI-Guided Inquiry Learning for Computational Drug Discovery	13	0	80	4-84	3-83	3-83	3-83	0-80
AIDD Commons	93	0	0	93	93	66	50	37
Build AI Drug Discovery Pipelines	93	42	0	43	42	34	15-34	0
ChEMBL Quick Tour	74	70	19	4-23	0-19	0-19	0-19	0-19
DeepChem Step-by-Step Tutorials	14	5	79	9-88	8-87	6-85	1-85	0-79
Advanced Machine Learning for Innovative Drug Discovery	20	0	73	0-73	0-73	0-73	0-73	0-73
Machine Learning for Drug Discovery and Development	18	0	75	0-75	0-75	0-75	0-75	0-75

Provider	Direct cells	Absent	Unknown	Explained	Worked	Executable	On real data	Assessed
O3DC Drug Discovery Benchmark Index	7	0	86	6-92	5-91	0-86	0-86	0-86
OpenMM Cookbook and Tutorials	79	75	14	3-17	2-16	2-16	2-16	0-14
Practical Cheminformatics Tutorials	23	15	70	8-78	8-78	6-76	3-76	0-70
RDKit Cookbook	49	45	44	3-47	3-47	0-44	0-44	0-44
TeachOpenCADD	93	41	0	46	40	32	23-32	23
Therapeutics Data Commons	15	0	78	8-86	7-85	6-84	5-84	0-78

Two rows in that table are worth a reader's attention. TeachOpenCADD is the only external provider with 93 direct cells and 0 unknown, because its whole surface was surveyed, and its lower bounds of 46, 40, 32, and 23 are therefore the strongest external lower bounds in the set. Build AI Drug Discovery Pipelines is the only other provider with 93 direct cells, and its 42 recorded absences are what lower its explained bound to 43. A provider surveyed at index level, such as one of the two benchmark hubs, carries a wide interval that measures the survey rather than the provider.

The two claims the audit was built to test both read as not established. The most comprehensive claim has a met bound condition, with an explained lower bound of 93 against a strongest external upper bound of 92, and it is not established because the gate's sixth condition, a public correction window, is not recorded. The most detailed claim has a met bound condition too, with a worked lower bound of 93 against a strongest external upper bound of 91, and 17 of the required 12 self-audit tracks containing executable or assessed coverage, 15 of them on real data, and it is not established for the same reason.

The audit also records what it did not find. Zero confirmed depth gaps have direct AIDD Commons evidence, and zero topics require self-audit before gap classification. Zero topics are below the executable threshold here while an external provider is already runnable there. Runnable coverage on real data is 43 topics here, against a largest external lower bound of 23 and a largest external upper bound of 86, so the unresolved cells still decide that comparison.

10.7 The limits of the audit

The twelve external systems are a frozen first comparison set and not an exhaustive multilingual census. The 618 unknown cells cannot support an absence claim or a promoted comparative claim. The executable threshold describes a directly visible runnable task surface, and it does not mean the audit reran every external notebook. External provider records have not been re-surveyed since 11 September 2026, so an external bound is a statement about what was known at that date and not about what a provider holds now. A comparator cell changes only through a source-backed correction.

The three v0.2b depth candidates were selected after passing all seven frozen eligibility criteria. Their selection fields are historical snapshots rather than current run status, and the selection-time text is preserved as history so that a reader can see what was believed at the time of selection and what was later delivered.

Chapter 11

What the independent reading found

11.1 The reading and its sandbox

The coverage numbers the project records about itself are written by the same hand that wrote the lessons. A reader who wants to know whether those numbers are inflated needs a second reader who did not write the material. That is what the independent reading is, and its design is the interesting part.

The reviewer was OpenAI Codex CLI 0.154.0 with the gpt-6-astra model at maximum reasoning effort. The self-scores it is compared with were assigned by Claude sessions, so the reading comes from a different model family than the scores. It is not a reading by a family that wrote none of the material, because Codex sessions also wrote parts of the course, and the report says so rather than claiming a blind replication that did not happen.

The first three batches ran in a read-only sandbox holding a git archive of one commit restricted to the lessons, lab guides, atlas pages, labs, resource records, taxonomy, and curriculum map. The curriculum's capability tags were removed, and the sandbox held no comparison records, no audit report, no roadmap, no self-score, and no Chinese edition. Web access was not used. The 149 completed shell commands across the three batches referenced no path outside the sandbox.

Later batches rebuilt the same sandbox for a new commit and re-read topics whose material had changed. Each of those batches checked its isolation by parsing every event as JSON rather than by searching for a path string, and each recorded the number of completed commands, the absence of an absolute path outside the sandbox, the absence of a parent-directory reference, and the absence of a web search. Four batches were run on 22 October 2026 by the same reviewer family and the same settings as the ten earlier ones, each in a sandbox holding only the published material and each with the real data snapshot present so that the command actually ran. Their isolation attestations record 17, 19, 22, and 17 completed commands.

The recorded reading also corrected three defects in its own prior text. The isolation string had double-counted the first three batches as 298 commands when the event streams show 149. It had said 34 for the fourth batch when the stream shows 17. And it had labelled two different batches as the sixth, where the second of those is the seventh. The sixth and seventh batches had produced readings for three topics and had carried no isolation attestation at all, which is now supplied.

The reading covered 93 topic units, the same set the audit scores.

11.2 The rubric

The rubric asks the reviewer to choose the deepest state the material demonstrably reaches for the topic as the taxonomy describes it, reading the passage rather than a title or a link, and to choose

the shallower state when the evidence is ambiguous. The executable state needs a runnable task in the repository about the topic with declared inputs and outputs. The assessed state needs an exercise with a worked answer, a rubric, or a verification target about the topic, and does not require anything runnable. Every executable or assessed reading states whether its runnable task rests on real data, on authored inputs, or on nothing runnable.

Because the audit scores depth ordinally, a recorded assessed state also claims executable. A reading of assessed with nothing runnable is therefore compared here as worked. That convention is what makes the comparison meaningful, and it is also the reason one topic's record did not move.

11.3 The result

Table 17 gives the two records at the five thresholds.

Table 17: The recorded depth and the independent reading at each threshold.

Threshold	Recorded here	Independent reading	Difference
explained	93	93	0
worked	93	93	0
executable	66	71	5
executable on real data	50	52	2
assessed	37	70	33

The reading agreed with the record on 60 topics, read 33 deeper and read 0 shallower. The validator refuses any record where it is shallower, so a topic this project claims more on than the reading supports cannot be published.

The assessed row is where the two records diverge most, and the reason is the rubric rather than a disagreement about the material. This project's rule is that a recorded assessed topic also claims executable, and the reading applies the same ordinal mapping in reverse, so a reading of assessed with nothing runnable is compared as worked. The reading therefore found far more assessed material than this project's record claims, because a written exercise with a worked answer satisfies the reading's assessed criterion whether or not it is accompanied by a runnable task.

11.4 The rule about the self record

Two rules govern what this project may do with the difference between the two records.

The first is that the self record may not claim more depth than the independent reading supports. Where the two disagree, the shallower value is recorded. A topic the reading scored as worked cannot be published as assessed, even though the project's own material plainly contains an exercise, because the reading is the instrument that checked it and the project is not the instrument that checks itself.

The second is that a topic cannot be raised on the strength of the reading alone. A raise means writing the missing material and running a new reading, not editing the record. Four topics below the worked threshold were raised in one round, and the audit records that the independent reading already allowed raising two of them without new material. That route was not taken, and the record says so. Each of the four was raised because a lesson was written and a new reading was run.

11.5 The topic the discipline refused to raise

One of the four new readings on 22 October 2026 re-read the topic that asks how a biologic construct's objectives are separated, and it came back as assessed with no runnable task for that topic. Under the ordinal mapping this compares as worked, so the topic's own record did not move.

The reading's reason is precise. The topic's written material explains and distinguishes all six objectives through a dornase alfa worked case and a HUMIRA exercise with complete answers, and it found no listed objective missing. The two neighbouring laboratories do not fill the gap for this topic. The SKEMPI lab teaches independent units and affinity evaluation, and the co-folding inspector classifies resource claims, and neither provides a runnable task addressing the six-objective distinction.

This is the discipline working, and it is worth stating as such rather than leaving it out. The reading found more material than the record claims in thirty-nine topics, and the obvious move would have been to raise the record to match. The rule forbids it, because the record is not raised by a number moving but by material arriving and a reading confirming it. In this topic the material for a runnable task has not been written, so the record stays where it is.

11.6 What the reading is not

The reading is an automated second reading of teaching depth. It is not reader evidence, not a human expert review, and not a measure of learning. The first three batches had no Python environment, so a note that local execution failed because RDKit was unavailable describes the sandbox. The first batches also excluded the scripts directory, so a note that an advertised checker script is missing describes the sandbox rather than the repository, where the script is published.

The reading also does not check the science. It checks whether the material reaches the depth the taxonomy describes. A lesson can reach assessed depth and still teach something the literature has moved past, and a laboratory can be protocol-valid and still answer a question no one has asked. The audit and the reading together bound what this project can say about its own coverage. They say nothing about whether the coverage is the coverage a reader needs.

Chapter 12

The current frontier method families

The curriculum now reads eight frontier method families, and each reading answers four questions about the family it covers. What the family is for, what the registry records establish about openness and licence, and where the evidence stops. Each is a reading of the registry records rather than of the systems themselves, and none of them runs a checkpoint or downloads a dataset.

12.1 Machine-learned interatomic potentials

A classical force field fixes the functional form of an energy surface and fits its parameters. A machine-learned interatomic potential lets the model learn the surface from data, and that substitution is the whole idea and the whole difficulty. A force field states in its own documentation which terms it contains and what was fitted to what, while a learned surface inherits the reach of whatever corpus produced it, and that corpus is often the part a reader cannot inspect.

The six released code bases the registry records for this family are NequIP, MACE, CHGNet, MatterSim, SevenNet, and TorchANI. NequIP is a code base for E(3)-equivariant machine-learned interatomic potentials that also distributes pretrained foundation potentials through its own model host (Batzner et al. 2022). MACE is a code base for machine-learned potentials with higher-order equivariant message passing, publishing pretrained potentials for materials and molecules, with an academic source licence on a subset (Batatia et al. 2023). CHGNet is a pretrained graph neural network potential for charge-informed atomistic modelling, trained on Materials Project relaxation trajectories. MatterSim is a deep learning atomistic model used as a machine-learning force field across the periodic table. SevenNet is a graph neural network interatomic potential package with pretrained universal potentials and a fine-tuning interface. TorchANI is a neural network potential library that registers built-in datasets.

Table 18 gives the openness of the six records. The pattern is the same in every row of the table and it is the point of the family.

Table 18: Openness of the six machine-learned interatomic potential records.

Record	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code
NequIP	available	available	available	available	unverified	unverified	unverified
MACE	available	available	available	available	unverified	available	available
CHGNet	available	available	unverified	available	unverified	available	unverified
MatterSim	available	available	available	available	unverified	unverified	unverified
SevenNet	available	available	unverified	available	unverified	unverified	unverified
TorchANI	available	available	available	available	unverified	unverified	unverified

Record	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code

A reader can therefore run these models and inspect them, and cannot reproduce their training from what is published. The data recipe is unverified in all six, and the evaluation code is open in one. A repository can contain the scripts that trained a model without containing a record of how the training set was assembled, and that is exactly the situation in this family.

The licences differ in kind. NequIP carries MIT over the source and CC BY 4.0 over the model weights, MACE carries MIT over the source and an Apache Software Licence over a subset of the weights, and MatterSim carries MIT over the source with the checkpoints under their own terms. CHGNet, SevenNet, and TorchANI carry MIT or BSD over the source, and for CHGNet and SevenNet the records state that no opened page gives a licence for the published checkpoints even though the repositories ship a licence for the code beside them. One case in this family illustrates why the same name can refer to two assets with different terms, since two assets share the OMat24 name and carry different licences. Another shows why two results are not independent, because one of the releases states that its core model is built on another's.

12.2 Single-cell foundation models and virtual cells

This family divides into two task contracts, and the gap between them is the teaching content. The first takes one cell as input and returns a vector, so a cell from a tissue, an assay, or a species the corpus never contained can still be placed in the same latent space, because the encoder needs no label and no fine-tuning. The second takes an intervention as input and returns a counterfactual, so a baseline cell state, a declared perturbation, and a context produce a predicted post-perturbation profile. A representation of a cell is not a prediction about an intervention.

The eight records are scGPT, scFoundation, CellFM, UCE, GEARS, CPA, PertFormer, and a zero-shot evaluation study. The scGPT repository describes a generative pretrained transformer for single-cell multi-omics that embeds gene expression into cell and gene representations (Cui et al. 2024). The scFoundation repository describes an asymmetric transformer pretrained on more than 50 million human single-cell transcriptomes. CellFM is a RetNet-based foundation model pretrained on transcriptomics of about 100 million human cells. UCE is a self-supervised model that maps single-cell transcriptomes into one shared biological latent space. GEARS is a graph-based deep learning model that predicts the transcriptional response to single and multi-gene perturbations by combining a gene embedding with a knowledge graph. CPA factorises single-cell transcriptional responses into basal, drug, and perturbation effects. PertFormer is a multimodal transformer with 3 billion parameters that predicts the effects of genetic, transcriptional, and sequence perturbations. The zero-shot study trains nothing.

Table 19 gives the openness of the eight records, and the weights column is the one that decides most downstream choices.

Table 19: Openness of the eight single-cell and virtual-cell records.

Record	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code
scGPT	available	unverified	available	available	unverified	unverified	unverified
scFoundation	available	unverified	available	available	available	unverified	unverified
CellFM	available	available	available	available	unverified	available	unverified
UCE	available	unverified	available	available	unverified	unverified	unverified
GEARS	available	unverified	unverified	available	unverified	available	available

Record	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code
CPA	available	unverified	available	available	unverified	unverified	unverified
PertFormer	available	unverified	available	available	unverified	unverified	unverified
Zero-shot evaluation	available	not applicable	not applicable	not applicable	unverified	not applicable	available

The licences attached to those weights range from a source-code-only MIT to a non-commercial, no-derivatives grant that forbids sharing an adapted model. A reader who plans to fine-tune, redistribute, or serve any of these models has to read the licence of the asset and not of the repository. scFoundation is the case in point, since its repository source is Apache-2.0 while its weights carry a separate licence granting use solely for non-commercial research purposes, and a reader who takes the Apache terms to the weights has taken the wrong licence. CellFM is the case in the other direction, since its repository and model card carry CC-BY-NC-ND-4.0 while its data archive on Zenodo is separately CC-BY-4.0.

The training corpora are the second limit. Five of the eight records leave the training corpus unverified. The scale figures are easy to read as coverage guarantees and they are not. The scGPT abstract reports more than 33 million cells, the scFoundation README gives more than 50 million human single-cell transcriptomes, the CellFM model card reports 19,914 samples holding 102,304,686 human cells, and the UCE paper reports an atlas of 36 million cells and more than 1,000 named cell types as a product of the model. Each figure bounds a quantity, and none is a cell-type inventory that can be checked against a named corpus, because the corpus is not named.

12.3 Protein and binder design

The protein and binder design family has records that are barely verified at all, and the reason is instructive. The registry holds ProteinMPNN, RFdiffusion, LigandMPNN, and Chroma as model records, and SAbDab2, the Observed Antibody Space, and the Immune Epitope Database as database records.

All four model records carry an unverified value in every one of the seven openness dimensions, and none of them carries a licence. The three database records carry a Creative Commons Attribution 4.0 licence, and only SAbDab2 establishes a training-data dimension. The family is therefore a case in which the curriculum teaches from sources whose asset status the registry has not yet established, and the honest statement of that is the teaching point.

The materials themselves are what a reader should study first. The ProteinMPNN repository describes a method for sequence design given a fixed backbone structure. The RFdiffusion repository describes a diffusion model for protein and binder design that generalises to tasks with little or no experimental data (Watson et al. 2023). BindCraft is a one-shot design method for protein binders that filters designs for predicted binding, monomericity, and AlphaFold confidence, and it reports a high experimental hit rate against seven targets (Pacesa et al. 2025). Chroma is a generative model over protein sequence and structure. The registry's own note for the family records that a well-defined benchmark, a comparison protocol, and a shared evaluation set do not yet exist for it, which is a statement about the field rather than about the registry.

A binder design method reports a hit rate against a set of targets, and a hit rate is a statement about a set. The way the method selects which target to try, which sequence variants to propose, and which filters to apply, all of it determines what the hit rate means, and all of it is a scientific claim that the registry records here do not establish. The family is also where the negative-selection lesson in the curriculum does its work, since a design that binds its target and

a decoy equally is a design whose specificity has not been tested.

12.4 Autonomous laboratory agents

An autonomous laboratory agent family contains three kinds of object, and they are not three versions of one thing. A planning agent operates on a chemistry objective in language plus the hardware and services it can call, and returns a plan, a sequence of typed calls, and a record of what each call returned. An instrument platform operates on the laboratory instrument and the protocol it runs, and returns instrument-driven experimental data under a software-controlled workflow. A capability benchmark operates on a fixed set of questions and a grading rule.

The Coscientist is recorded as an autonomous chemistry research agent that plans and executes multi-step experiments with laboratory hardware and language models (Boiko et al. 2023). The Robin record describes a multi-agent system for automated small-molecule discovery that runs search, design, and analysis in one loop. The Agent Laboratory record describes a workflow that plans, executes, and documents a machine-learning research project end to end. The ChemCrow record describes a system that augments a language model with chemistry tools (Bran et al. 2024). The AutoLabs record describes an open autonomous laboratory platform for running instrument-driven experiments under a software-controlled workflow. The AI-Researcher record describes a research-agent system and its benchmark suite. The ChemBench and LAB-Bench records describe the two benchmarks that ask whether a model can do chemistry at all (Mirza et al. 2024; Laurent et al. 2024).

For this family the released artefact is the system rather than the protocol, and the records say so in their own words. The Robin record states that the released artefact is the system and not a frozen evaluation with a protocol the registry can inspect. The Agent Laboratory record adds that the reported results depend on the services and the models available at the time, and that the registry does not establish that a reported run can be reproduced. The Coscientist record adds that the published evaluation depends on hardware the project operated, and that the record does not establish that the result can be reproduced on other hardware.

The licences carry the same shape. The Coscientist licence is Apache-2.0 with a Commons Clause, and its text states that the grant does not include the right to sell the software. The ChemCrow licence does not cover the hardware, the chemistry services, or any third-party data the agent calls. The AutoLabs licence is BSD-2-Clause over the platform source and does not cover the instruments, the facility, or any consumable. One record states no licence at all, since the AI-Researcher repository root carries no licence file and its README states no licence. A reader who writes that this suite is MIT-licensed, or open source, or freely reusable has added a term that no opened file states.

One practical detail belongs with the AutoLabs record. Its default branch is master rather than main, so a reader who follows a main-branch path finds no licence file. The licence text is reachable at the branch the project uses.

12.5 The ADMET and DMPK tool frontier

The ADMET and DMPK family exists in this curriculum because the field's tooling has consolidated into platforms that cover many endpoints at once, and because that consolidation changes what a reader has to check. The five records are ADMET-AI, ADMETlab, Polaris, TDC, and OpenADMET Models.

ADMET-AI is a machine-learning platform for the evaluation of ADMET properties over large chemical libraries, and its record is the most completely evidenced in the family, with all seven openness dimensions available (Swanson et al. 2024). ADMETlab 3.0 is an online platform pre-

dicting physicochemical, ADME, and toxicity properties, built on DMPNN models, with an API and decision support (Fu et al. 2024). Polaris is a hosted platform for sharing drug-discovery datasets and benchmarks, with a Python client that downloads datasets and loads benchmark splits. TDC is the task hub described in Chapter 3, and its record is unverified in every dimension. OpenADMET Models is a set of model implementations and training routines for building and testing ADMET predictors.

The licenses do not follow the openness pattern, and the two records where they diverge most are instructive. ADMETlab is a hosted web platform whose repository carries a GNU Affero General Public License version 3, and its weights carry a New BSD term, and its access level is source available rather than open source. Polaris carries an Apache-2.0 term over the source and a separate platform agreement over the service, and its weights are unverified, so the terms of the model assets a reader downloads through the client are not established by the record. The TDC catalogue carries a blanket "Not Specified, CC BY 4.0" tag applied identically across its entire catalogue, and the registry's own caveat records that this applies to a data release that was separately checked and found to sit behind closed-access terms, which is the case in which a licence tag does not establish the terms of the underlying data.

The two benchmark hubs are in the family for a reason that goes beyond convenience. A benchmark hub that lists a task, and a system that teaches the task, deliver different things, and the audit of Chapter 10 scores them as different classes for exactly this reason.

12.6 Flow matching and GFlowNet objectives

A generative model has a target distribution, and the training rule decides which one the sampler converges to. Under a maximum-likelihood objective the target is the data distribution, so the sampler reproduces the corpus together with the regions the corpus barely covers. Under a flow-matching objective the target is still the data distribution, reached by transporting a simple base distribution along a chosen probability path, so the path decides what the intermediate states are and how many steps the solver needs, but not the distribution the sampler ends at (Lipman et al. 2022). Under a GFlowNet objective the target is proportional to a reward, so the sampler concentrates on the states the reward prefers whether or not the corpus contains many of them (Bengio et al. 2021).

Two consequences follow for anyone reading a candidate list. A maximum-likelihood sampler and a reward-proportional sampler visit different parts of the space, so their outputs are not two draws from one population and cannot be pooled or compared without naming the target. And the validity of a reward-proportional sampler rests entirely on the reward being a faithful proxy, because nothing in the objective ties a high-reward state to an object that exists, is stable, or can be made.

The thirteen records in this family are the three library records and the ten model records that use them. The libraries are GFNOrg's gflownet, torchgfn, and the flow-matching library, and they are the parts of the family a reader can execute. The models are FlowMol3, FragFM, DeFoG, MolFM, InstaFlow, AGFN, RxnFlow, TacoGFN, CGFlow, and Graph-GRPO.

Table 20 gives the openness of the thirteen records, and the weights column is the sharpest boundary in the family.

Table 20: Openness of the thirteen flow-matching and GFlowNet records.

Record	Inference code	Training code	Weights	Architecture	Data recipe	Training data	Evaluation code
flow_matching library	available	available	not available	unverified	unverified	unverified	unverified
GFNOrg gflownet	available	available	not available	unverified	unverified	unverified	available
torchgfn	available	available	not available	unverified	unverified	unverified	available
FlowMol3	available	available	available	unverified	unverified	available	unverified
FragFM	available	available	available	unverified	unverified	unverified	unverified
DeFoG	available	available	available	unverified	unverified	unverified	unverified
MolFM	available	available	not available	unverified	unverified	unverified	unverified
InstaFlow	available	available	available	unverified	unverified	unverified	unverified
AGFN	available	available	available	unverified	unverified	unverified	unverified
RxnFlow	available	available	not available	unverified	unverified	unverified	unverified
TacoGFN	available	available	available	unverified	unverified	unverified	unverified
CGFlow	available	available	available	unverified	unverified	unverified	unverified
Graph-GRPO	available	available	available	unverified	unverified	unverified	unverified

Three of the thirteen records carry no weights, and they are the three libraries, where the point is the training procedure rather than a released checkpoint. The flow-matching library carries a Creative Commons non-commercial grant, which is the most restrictive term in the family, and the two GFlowNet libraries carry MIT and Apache-2.0. The model records are overwhelmingly MIT, with InstaFlow adding a Creative Commons non-commercial term and Graph-GRPO adding an Apache-2.0 term.

The architecture column is unverified for all thirteen, and that is a limitation of the registry rather than of the field, since every one of these systems states its objective in its own paper. What the records do not establish is whether the published architecture description is complete enough to reimplement the training rule, and for a family whose distinguishing feature is the training rule, that is the dimension a reader would want.

12.7 Nucleic-acid and peptide design models

The nucleic-acid and peptide family is smaller than the others and its records are more complete. The five records are RNA-FM, RibonanzaNet, PepHAR, HELM-BERT, and ProteinFlow. RNA-FM is a language model that encodes RNA sequences as nucleotide embeddings, pretrained by masked-token prediction on 23.7 million non-coding RNA sequences. RibonanzaNet is a network consolidated from the Ribonanza crowdsourced chemical-mapping data and fine-tuned for secondary structure. PepHAR samples hotspot residues with an energy-based density model and then extends peptide fragments autoregressively while constraining the peptide geometry. HELM-BERT is an encoder-only transformer over peptides written in HELM notation. ProteinFlow is a pipeline that selects protein structures from the PDB or SAbDab, computes structural annotations, and clusters chains by sequence or CDR similarity.

The licensing in this family is uniform. All five model records carry MIT over the repository source, and RNA-FM adds an Apache-2.0 term and ProteinFlow carries BSD-3-Clause. The uni-

formity is itself the useful fact, because it means a reader comparing two of these systems is not comparing two sets of terms.

The one record where the terms are not uniform is HELM-BERT, whose weights are hosted separately and whose licence terms the record states as separate from the repository's. The two licence questions in this family that a reader has to answer are not about the code. They are about which corpus each model was trained on and whether that corpus can be used, since the peptide and nucleotide design field draws its training data from competition and community collection efforts with their own terms, and the ProteinFlow record in particular documents a precomputed release whose split setting is given with two different values in different contexts.

12.8 Reading a structure from an image

The last family is the smallest and the most recent. Six records cover the task of predicting a molecular structure from a picture of one, and the lesson that reads them was written because the field's tooling had consolidated and the registry had not kept up with the distinctions the tools make.

The DECIMER Image Transformer is a vision transformer trained to interpret a molecular image (Rajan et al. 2023). The DECIMER.ai platform is an open platform for automated optical chemical structure identification, segmentation, and recognition in scientific publications, and it is the piece a reader reaches for when the image is embedded in a paper rather than supplied as a clean rendering (Rajan et al. 2023). MolScribe is a model that predicts atoms and bonds directly from an image and generates the graph, which is a different approach from recognising characters (Qian et al. 2023). The MolNexTR model predicts the atoms and bonds of a molecular image and converts the prediction to a MolFile and SMILES, using a ConvNeXt and a vision transformer. SwinOCSR uses a Swin Transformer backbone with a Transformer decoder to turn images of chemical structures into SMILES. The Img2Mol model predicts a molecule from a picture of a chemical structure, and its repository ships the inference implementation, two example notebooks, and a downloadable set.

The spread in the openness records is the reason the family needed a reading.

The two records where training code is recorded as not available are the two where the model is distributed as a hosted service or a released checkpoint, and the two where every dimension is unverified are the two whose records have not yet been re-reviewed. The licences carry a similar spread. MolNexTR carries Apache-2.0 over the source and CC BY 4.0 over the data, MolScribe and DECIMER carry MIT over the source, and Img2Mol carries Apache-2.0 over the source and a Creative Commons non-commercial term over the model, which is the most restrictive term among the six and the one that most constrains a commercial use.

The reading states where the evidence stops with particular clarity here, because optical structure recognition is a field where an impressive demo and a usable method are far apart. Every one of these systems reports an accuracy on a benchmark, and every one of them reports a training set that is a mixture of rendered images and images cropped from published documents, and the two are not the same distribution. A model that scores well on rendered images and poorly on a document crop has not failed at optical structure recognition, it has failed at the part of the task that the benchmark does not contain.

12.9 What the eight readings establish together

Three conclusions hold across the eight readings, and they are the conclusions the curriculum now teaches. First, the openness a reader can act on is not the openness of the repository, and a repository that opens its code can still withhold the corpus, the data recipe, or the evaluation

script. Second, a licence is a property of an asset and a family can hold records whose licences differ in kind even when their openness records look alike, which is why the registry records the scope of each licence beside its name. Third, every one of these readings stops at the registry, and none of them ran a checkpoint, downloaded a dataset, or measured a performance number, so none of them is evidence about how any of these systems performs.

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